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Tracking Deep Lithospheric Events with Garnet-Websterite Xenoliths from Southeastern Australia

Jianggu Lu[1,2]*, William L. Griffin[1], Romain Tilhac[1], Qing Xiong[1,2], Jianping Zheng[1,2] and Suzanne Y. O'Reilly[1]

[1]Australian Research Council Centre of Excellence for Core to Crust Fluid System (CCFS) and GEMOC, Department of Earth and Planetary Sciences, Macquarie University, Sydney, NSW 2109, Australia; [2]State Key Laboratory of Geological Processes and Mineral Resources, School of Earth Sciences, China University of Geosciences, Wuhan 430074, China

*Corresponding author. E-mail: jianggu.lu@mq.edu.au

ABSTRACT

Pyroxenites provide important information on mantle heterogeneity and can be used to trace mantle evolution. New major and trace element and Sr-, Nd-, and Hf-isotope analyses of minerals and whole-rock samples of garnet websterites entrained in basanite tuffs in Bullenmerri and Gnotuk maars, southeastern Australia, are here combined with detailed petrographic observations to constrain the sources and genesis of the pyroxenites, and to trace the dynamic evolution of the lithospheric mantle. Most garnet websterites have high MgO and Cr_2O_3 contents, relatively flat light rare earth element (LREE) patterns ($[La/Nd]_{CN} = 0{\cdot}77$–$2{\cdot}22$) and ocean island basalt-like Sr-, Nd-, and Hf-isotope compositions [$^{87}Sr/^{86}Sr = 0{\cdot}70412$–$0{\cdot}70657$; $\varepsilon_{Nd(t)} = -0{\cdot}32$ to $+4{\cdot}46$; $\varepsilon_{Hf(t)} = +1{\cdot}69$ to $+18{\cdot}6$] in clinopyroxenes. Some samples show subduction-related signatures with strong enrichments in large ion lithophile elements and LREE, and negative anomalies in high field strength elements, as well as high $^{87}Sr/^{86}Sr$ (up to 0·709), and decoupled Hf- and Nd-isotope compositions [$\varepsilon_{Nd(t)} = -3{\cdot}28$; $\varepsilon_{Hf(t)} = +11{\cdot}6$). These data suggest that the garnet pyroxenites represent early crystallization products of mafic melts derived from a convective mantle wedge. Hf model ages and Sm–Nd mineral isochrons suggest that these pyroxenites record at least two stages of evolution. The initial formation stage corresponds to the Paleozoic subduction of the proto-Pacific plate beneath southeastern Australia, which generated hydrous tholeiitic melts that crystallized clinopyroxene-dominated pyroxenites at ~1420–1450°C and ~75 km depth in the mantle wedge. The second stage corresponds to Eocene (*c*. 40 Ma) back-arc lithospheric extension, which led to uplift of the former mantle-wedge domain to 40–60 km depths, and subsequent cooling to the ambient geotherm (~950–1100°C). Extensive exsolution and recrystallization of garnet and orthopyroxene (± ilmenite) from clinopyroxene megacrysts accompanied this stage. The timing of these mantle events coincides with vertical tectonism in the overlying crust.

Key words: mantle garnet pyroxenite xenoliths; hydrous tholeiitic melts; mantle-wedge melts; high-pressure pyroxenite exsolution; southeastern Australia mantle evolution

INTRODUCTION

Studies of ultramafic xenoliths and massifs have provided much information on the Earth's upper mantle. In a broad sense, these mantle-derived rocks provide direct evidence that the subcontinental upper mantle consists dominantly of peridotite with minor pyroxenite (<10%; e.g. Downes, 2007; Bodinier & Godard, 2014). Consequently, nearly all petrological studies of mantle melting traditionally start with the assumption that primary mantle melts equilibrate with olivine. More recently, the role of pyroxenites in the generation of mantle-derived magmas has been increasingly considered (Hirschmann & Stolper, 1996; Sobolev *et al.*, 2005; Tuff *et al.*, 2005). Preferential melting of pyroxenite (or eclogite) layers or veins accounts satisfactorily for isotopic variations in oceanic basalts, and potentially for isotopic discrepancies between mid-ocean ridge basalts (MORB) and residual abyssal peridotites (Salters & Dick, 2002). Experimental petrology and isotopic considerations suggest that even less than 2% of pyroxenite in basalt source regions may result in significant enhancements of magma production and may be responsible for a sizeable fraction of MORB melts (Pertermann & Hirschmann, 2003*a*). Moreover, pyroxenites may represent the missing link between the average andesitic composition of the continental crust and the basic primary arc magmas (e.g. Kelemen, 1995; Rudnick, 1995; Hawkesworth & Kemp, 2006; Lee *et al.*, 2006, 2007; Xiong *et al.*, 2014; Lee & Anderson, 2015; Tilhac *et al.*, 2016). Mantle pyroxenites may thus play a key role in global geodynamic processes, although they represent only a small volume of the upper mantle, and a better understanding of their distribution and origin is critical to constraining the evolution of geochemical heterogeneities of mantle through time.

Pyroxenites occur as minor layers and lenses in peridotite massifs (Kornprobst *et al.*, 1990; Pearson *et al.*, 1993; Garrido & Bodinier, 1999; Takazawa *et al.*, 1999; Gysi *et al.*, 2011) and as xenoliths in kimberlites (e.g. Roden *et al.*, 2006; Aulbach *et al.*, 2007; Gonzaga *et al.*, 2010; Smit *et al.*, 2014; Aulbach & Jacob, 2016; Newton *et al.*, 2016) and alkali basalts (e.g. Griffin *et al.*, 1984, 1988; Xu *et al.*, 1996, 1998; Xu, 2002; Ishikawa *et al.*, 2004; Bizimis *et al.*, 2005; Yu *et al.*, 2010; Ackerman *et al.*, 2012). The petrogenesis of these pyroxenites is controversial (Downes, 2007; Bodinier & Godard, 2014). Most are interpreted as high-pressure cumulates that crystallized from a variety of melts migrating through the mantle (Frey, 1980; Irving, 1980; Medaris & Syada, 1999; Fabries *et al.*, 2001; Xu, 2002; Upton *et al.*, 2003; Ishikawa *et al.*, 2004; Bizimis *et al.*, 2005; Berger *et al.*, 2007; Dantas *et al.*, 2007; Medaris *et al.*, 2013; Xiong *et al.*, 2014; Martin *et al.*, 2015), or formed by metamorphism of subducted oceanic crust tectonically incorporated into the mantle (Allègre & Turcotte, 1986; Kornprobst *et al.*, 1990; Morishita *et al.*, 2003; Obata *et al.*, 2006; Yu *et al.*, 2010; Svojtka *et al.*, 2016). Melt–rock and related peridotite-replacement reactions also have been recognized as a viable model for the origin of some mantle pyroxenites (Garrido & Bodinier, 1999; Liu *et al.*, 2005; Bodinier *et al.*, 2008; Tilhac *et al.*, 2016, 2017). Abundant pyroxenite xenoliths are found in the Mesozoic–Cenozoic basalts of southeastern Australia and they have been interpreted as clinopyroxene-dominant cumulates, modified by exsolution and recrystallization during cooling to the ambient geotherm (Irving, 1974; Wilkinson & Kalocsai, 1974; Griffin *et al.*, 1984; Stolz, 1984; Lu *et al.*, 2017). Most previous studies mainly focused on the reconstruction of a paleo-geotherm for southeastern Australia (SEA geotherm; i.e. Griffin *et al.*, 1984; O'Reilly & Griffin, 1985, 1996; O'Reilly, 1989; O'Reilly *et al.*, 1997), and there are few constraints on the source of the pyroxenites, or the conditions and timing of their formation.

Therefore, this study has re-examined garnet pyroxenites from the Quaternary maars at Lakes Bullenmerri and Gnotuk (Victoria, southeastern Australia), using new geochemical analytical and imaging technologies, and integrates petrological and major and trace element compositions of minerals and whole-rock samples with Sr-, Nd-, and Hf-isotope compositions. The aims are to better constrain the origin of the parental magmas to the garnet pyroxenites and to determine the dynamic processes that trigger the melt generation and pyroxenite formation.

GEOLOGICAL SETTING AND TECTONIC CONTEXT

The Australian continent consists of two broad tectonic regimes: the western two-thirds, which comprises dominantly Archean–Proterozoic cratonic terranes overlain in places by Paleozoic cover, and the eastern third consisting of the complex Paleozoic–Mesozoic Tasmanides, a collage of several orogenic belts and an internal Permian–Triassic foreland fold–thrust belt (Fig. 1a). The Tasman Line is the boundary of these two domains and it traces the locus of the ~600 Ma rifting margin (the younger of two rifting events) during the break-up of Rodinia (Veever, 1984). A series of convergent-margin orogenic belts subsequently docked with cratonic Australia from ~540 to 200 Ma. Successive SW to NE accretion events formed the southern Tasmanides, now represented by the Delamerian, Lachlan, Thomson and New England fold belts (Fig. 1a; Glen, 2005). The Lachlan and Delamerian fold belts (Fig. 1b) crop out in western Victoria, separated by a major crustal feature, the Moyston Fault (VandenBerg *et al.*, 2000; Cayley *et al.*, 2011).

The Delamerian Orogen formed a west-vergent foreland-type fold-and-thrust belt that overprinted the Neoproterozoic Adelaidean successions from *c.* 514 to 490 Ma (Betts *et al.*, 2002; Foden *et al.*, 2006). The Lachlan fold belt is considered to be a wide (~750 km), early to mid-Paleozoic continental margin with repetitive rifting, collision, and subduction–accretion episodes associated with granitic magmatism and basaltic–andesitic volcanism from *c.* 440 to 340 Ma (Coney, 1992; Foster & Gray, 2000). During the Mesozoic and the Cenozoic, Australia moved northward from Antarctica, forming a series of rift basins (i.e. Otway Basin in western Victoria) along the southern seaboard of the Australian continent (Fig. 1b). The Otway Basin is an east–west trough initiated early in the Mesozoic period and contains a thick sequence of Mesozoic and Cenozoic sedimentary and volcanic rocks (Edwards *et al.*, 1996; Holdgate & Gallagher, 2003; Briguglio *et al.*, 2015). Basaltic lavas, tuffs and other pyroclastic flows are widespread in the Otway Basin. Many are Pleistocene to Recent in age, and are referred to as the 'Newer Volcanic Province' (NVP). The NVP can be subdivided into two series, the early Plains (dominant) and the slightly later Cones, on the basis of

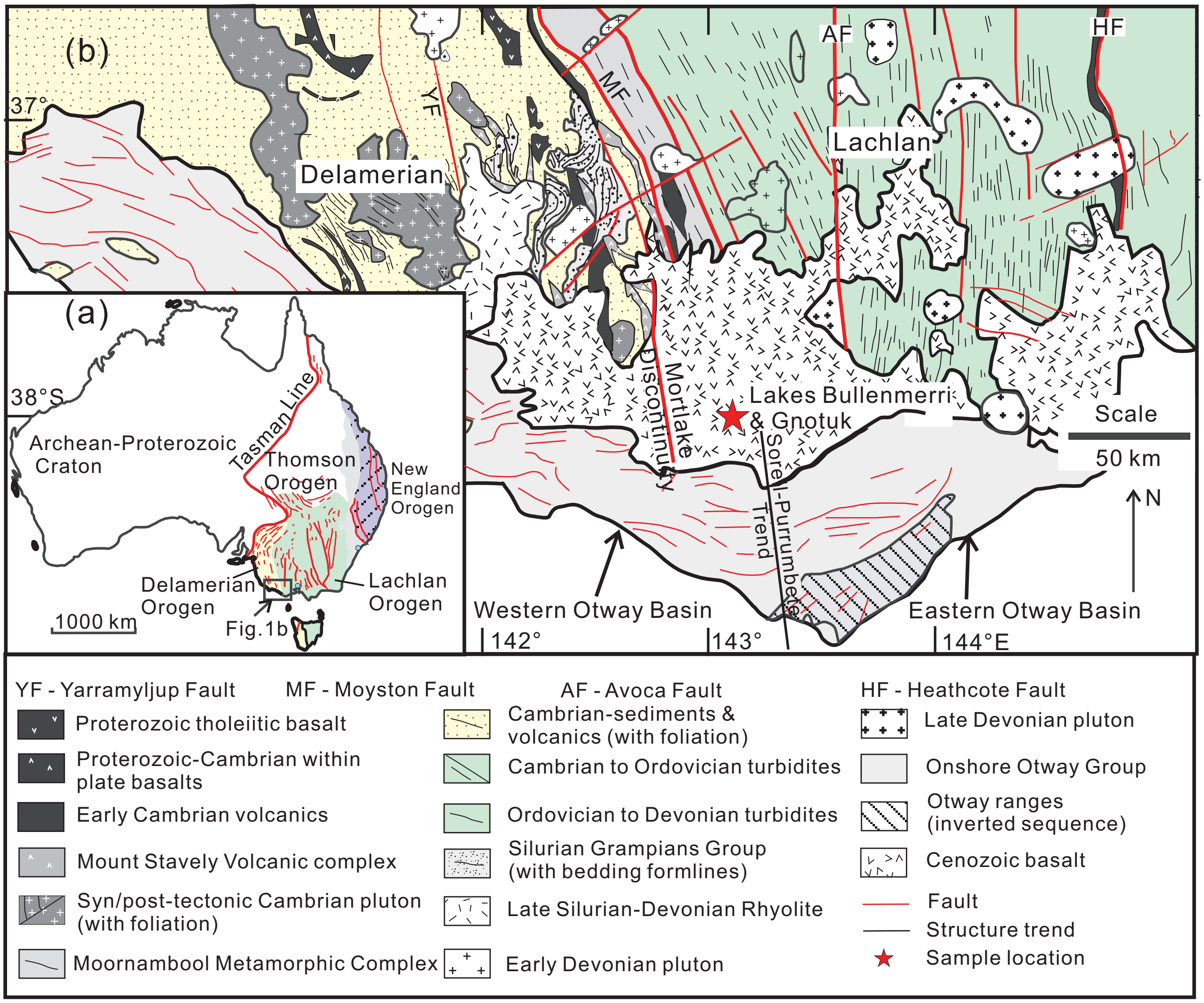

Fig. 1. (a) Major tectonic divisions of Australia. Area west of the Tasman Line consists of Archean–Proterozoic cratonic terranes; the eastern part consists of the Paleozoic–Mesozoic Tasmanides (Delamerian, Lachlan, New England, and Thomson orogenic belts. (b) Simplified geological map of western Victoria showing the Lachlan orogenic belt, Delamerian orogenic belt and Otway Basin (compiled from Miller *et al.*, 2002, 2005).

geomorphology and composition (Wellman & McDougall, 1974). The Plains basalts are mainly tholeiitic to transitional in composition, whereas the Cones comprise silica-undersaturated alkali basalts, commonly erupted as scoria or cinder cones, maars and tuff rings; they are locally very rich in mantle-derived xenoliths (Edwards *et al.*, 1996).

The mantle xenoliths used in this study are from Lakes Bullenmerri and Gnotuk (Bullenmerri–Gnotuk), adjacent vents, now maars, <1 km apart (Fig. 1b). The locality is close to the Mortlake discontinuity, defined in the lithospheric mantle by a contrast in the Sr- and Pb-isotopic compositions of NVP Plains basalts; $^{87}Sr/^{86}Sr$ is low to the west and high to the east (Price *et al.*, 1997). The Re–Os age distribution of the mantle xenoliths also suggests a significant age change from Proterozoic in the west to dominantly Phanerozoic to the east of the Mortlake discontinuity (Handler *et al.*, 1997; Handler & Bennett, 2001). The discontinuity appears to be coincident with the surface expression of the east-dipping Moyston Fault, which separates the Paleozoic Delamerian–Lachlan fold belts (Glen, 1992). The basanite tuffs have compositions within the range of the NVP Cones basalts, with low MgO (~6·71 wt %) and CaO (~6·89 wt %) and high alkali contents; they also have limited ranges in Sr and Nd isotope compositions ($^{87}Sr/^{86}Sr = 0{\cdot}7039$–$0{\cdot}7041$, $^{143}Nd/^{144}Nd = \sim 0{\cdot}51283$; Griffin *et al.*, 1988; Stolz & Davies, 1988).

PETROGRAPHY

The pyroxenite suite, including both spinel- and garnet-bearing pyroxenites, is clearly distinguished from the peridotite suite based on modal abundances, pyroxene compositions and microstructures. Pyroxenites occur separately, or as layers and pods in spinel lherzolite xenoliths, suggesting that they were intrusive into the lherzolites (Griffin *et al.*, 1984). Although Griffin *et al.*

(1984) reported the presence of garnet-free pyroxenites in the Bullenmerri–Gnotuk suite, all the samples studied here contain garnet. Most xenoliths are small (~5 cm in the longest dimension), but some samples reach 20 cm. They may be sub-spherical or ovoid but more commonly are angular in shape with rectangular to triangular cross-sections. Among the nine garnet websterites and one garnet clinopyroxenite chosen for this study, a wide range of microstructural types has been observed, from coarse to porphyroblastic (Fig. 2; Table 1).

Porphyroblastic textures

Most samples show porphyroblastic textures with large pyroxene megacrysts. Megacrysts (Fig. 2a and b) dominantly are clinopyroxene (Cpx) with exsolved lamellae of garnet (Grt) + orthopyroxene (Opx) ± ilmenite (Ilm) (Fig. 3a and b); minor Opx megacrysts have exsolution lamellae of Cpx and Grt (Fig. 3c). Amphibole (Amp) also occurs as lamellae and pods in clinopyroxene megacrysts and replaces garnet and orthopyroxene lamellae (Fig. 3d). The matrix shows subsolidus recrystallization textures with an equigranular mosaic of fine-grained (0·5–2 mm) Cpx + Grt + Opx + Amp (Fig. 3e). Cpx and Opx neoblasts are generally free of exsolution and well equilibrated in the texture (Fig. 3e). Discrete garnet grains are polygonal and are locally aligned among the randomly distributed Cpx subgrains and neoblasts, indicating that they probably represent former lamellae exsolved from original coarse Cpx, the grain size of which was reduced to fine neocrysts with increasing degree of recrystallization (Lu *et al.*, 2017). Amphibole is pervasive in the recrystallized assemblages and occurs as subhedral grains in apparent textural equilibrium with the anhydrous phases. Some samples (e.g. DR9748) also contain amphibole veins that crosscut the sample along a set of cracks (Fig. 3f) and locally enclose spinel grains with strongly embayed margins (Fig. 3g). Textural observations indicate at least two episodes of amphibole formation, as the veined amphibole postdates the discrete (pervasive) amphibole in the recrystallized areas. The transition from lamellar to mosaic microstructure is not sharp (Fig. 3h–i), reflecting varying degrees of recrystallization within individual thin sections. Detailed descriptions of the microstructures have been given by Lu (2018).

Coarse textures

In coarse microstructures, garnets are up to 5 mm across and are set in slightly finer granoblastic Cpx, Opx and Amp (Figs 2c and 4a, b). Coarse-grained garnets are polygonal or rounded, and partially or completely replaced by an extremely fine-grained, symplectitic aggregate (kelyphite) consisting of Opx, spinel (Spl) and Cpx. Rare Grt aggregates also contain small round (0·1–0·5 mm) Spl grains (Fig. 4c) and accompany the intergrowths of vermicular Spl and Opx (Fig. 4d), probably representing the breakdown of Grt lamellae exsolved from Cpx megacrysts under different *P–T* conditions. In sample GN35, for instance, the intergrowth of Opx and Spl is parallel to the Opx and Grt exsolution lamellae in the Cpx megacryst (Fig. 2b), suggesting that the sample shows two apparently sequential reactions: (1) Cpx I → Cpx II + Grt I ± Opx I; (2) Grt I → Grt II + Spl + Opx II. One sample (BM96-3) contains phlogopite (Phl)-bearing pockets in which subhedral Phl grains are surrounded by open cavities associated with intergrowths of fine-grained Cpx, Spl and Opx (Fig. 4e and f). Phl is in direct contact with the neighboring Cpx and Opx but is separated from coarse Grt by fine-grained Cpx + Opx + Spl, indicating that

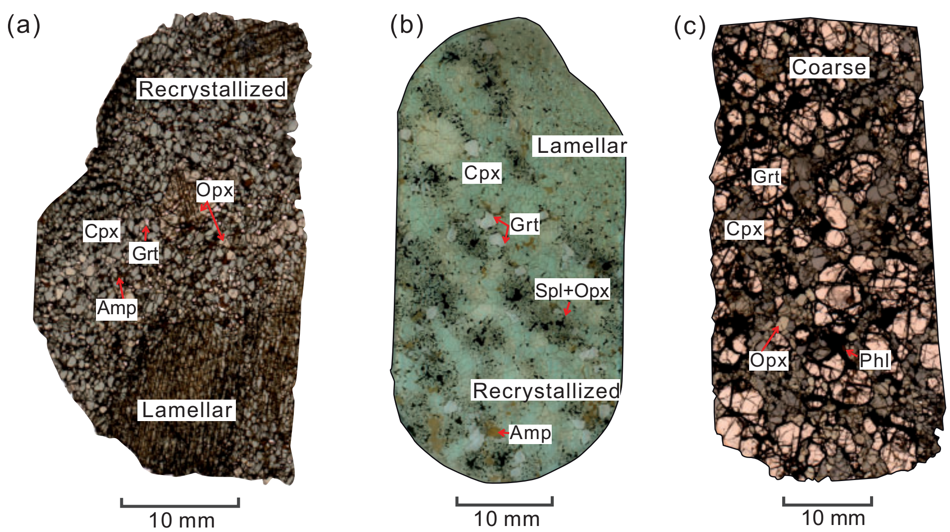

Fig. 2. Scanned images of thin sections of garnet websterites. (a) BM99-4 shows Cpx and Opx megacrysts and subsolidus recrystallization textures. (b) GN35 shows intergrowths of spinel and orthopyroxene parallel to exsolution lamellae of fine-grained garnet and orthopyroxene in the Cpx megacryst. (c) BM96-3 shows coarse polygonal Grt set in a matrix of recrystallized Cpx and Opx; Phl-bearing pockets surrounding coarse Grt should be noted. Cpx, clinopyroxene; Opx, orthopyroxene; Grt, garnet; Phl, phlogopite; Amp, amphibole; Spl, spinel.

Table 1: Summary of petrological description and calculated modal compositions (wt %) for garnet pyroxenites in Lakes Bullenmerri and Gnotuk

Sample no.	Rock type	Texture	Mg#	*T–P* estimates		Estimated modal abundances (wt %)‡						
				Primary*	Re-equilibration†	Cpx	Opx	Grt	Amp	Spl	Ilm	Phl
BM96-2	Grt websterite	Coarse	78·2		1094°C, 1·8 GPa	52	13	30	4	1		
BM99-6	Grt clinopyroxenite	Coarse	84·4		1060°C, 1·6 GPa	61		38	<0·5	1		
BM99-4	Grt websterite	Porphyroblastic	85·1	1450°C, 2·3 GPa	982°C, 1·5 GPa	57	20	18	5			
BME-1	Grt websterite	Porphyroblastic	82·3	1420°C, 2·4 GPa	1011°C, 1·4 GPa	55	11	33	<0·5			
BM99-2	Grt websterite	Porphyroblastic	80·7	1450°C, 2·4 GPa	1064°C, 1·7 GPa	55	13	31				
BM96-3	Grt websterite	Coarse	73·6		1010°C, 1·4 GPa	23	15	60	<0·5			2
BM99-5	Grt websterite	Porphyroblastic	76·1	1460°C, 3·0 GPa	1047°C, 1·7 GPa	64	28	5	3		<0·5	
DR9748§	Grt websterite	Granoblastic	85·2		1045°C, 1·6 GPa	40(41)	44(45)	11(13)	5(0)			
DR10165§	Grt websterite	Porphyroblastic	83·2		970°C, 1·2 GPa	49(46)	25(28)	17(17)	3(0)	6(8)		
GN35§	Grt websterite	Porphyroblastic	78·4		1063°C, 1·6 GPa	54(50)	27(37)	8(0·28)	4(0)	7(12)		

*CaO–MgO–Al_2O_3–SiO_2 (CMAS) system from Gasparik (2014).
†Ellis & Green (1979) Cpx–Grt thermometer and Wood (1974) Al-in-Opx barometer.
‡Mineral modes were estimated from point-counted modes and converted to wt % from mineral densities using $\rho_{Cpx} = 0{\cdot}32$, $\rho_{Opx} = 0{\cdot}32$, $\rho_{Grt} = 3{\cdot}71$, $\rho_{Amp} = 3{\cdot}12$, $\rho_{Spl} = 3{\cdot}6$, $\rho_{Ilm} = 4{\cdot}72$, $\rho_{Phl} = 2{\cdot}8$. Least-squares fitted modes (results in parentheses) were calculated from whole-rock and mineral major element compositions.
§Samples are from Griffin *et al.* (1984, 1988) and O'Reilly & Griffin (1985).

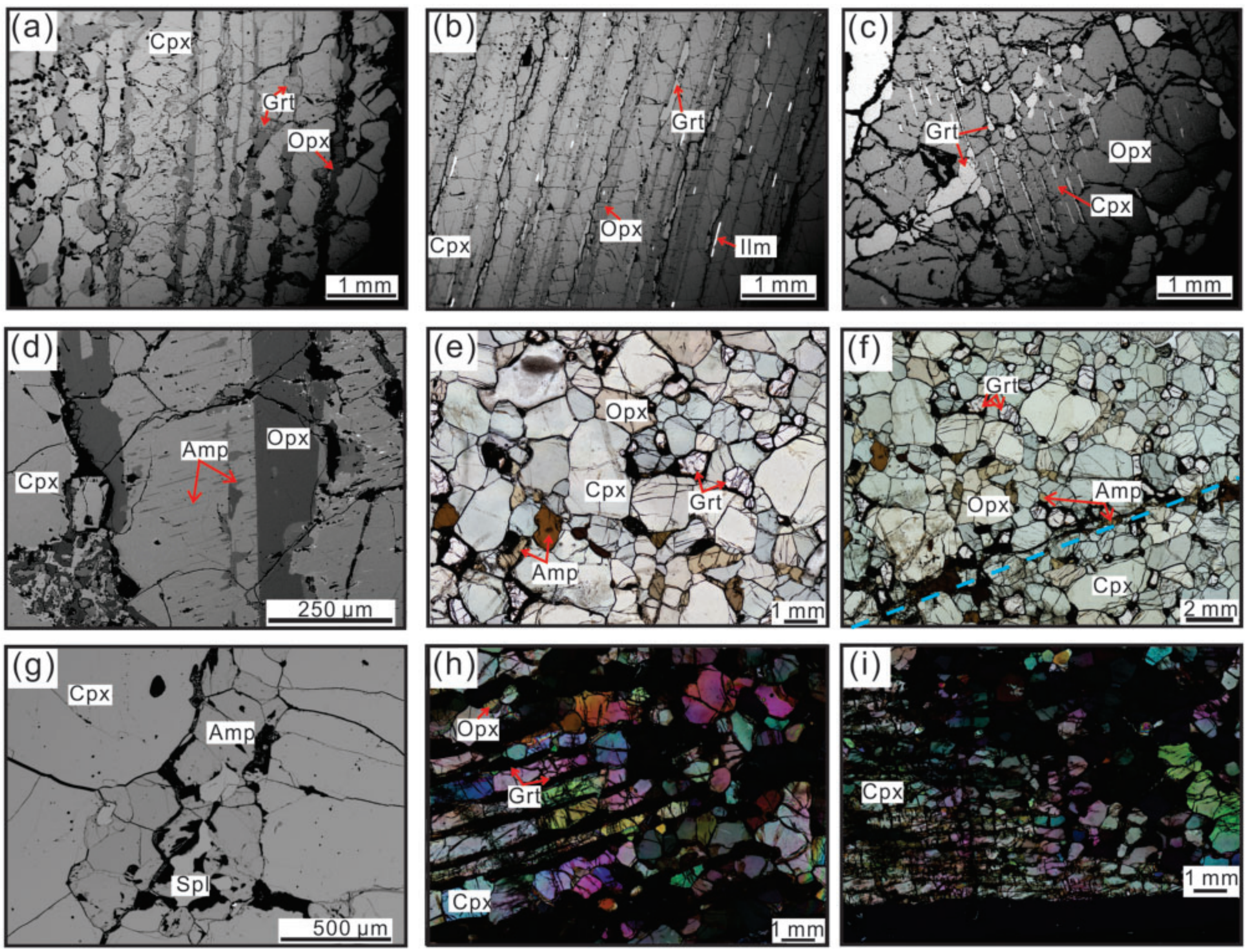

Fig. 3. Photomicrographs [a–d and g, back-scattered electron (BSE) images; e, f, plane-polarized light photographs; h, i, cross-polarized light photographs] showing porphyroblastic textures of garnet pyroxenite xenoliths (a, c and i, BM99-4; b, BM99-5; d, BME-1; e, f, DR9748; g, BME-1; h, BM99-2) from Lakes Bullenmerri and Gnotuk, southeastern Australia. (a) Cpx megacryst with Opx lamellae and Grt blebs; (b) Cpx megacryst with Opx, Grt and ilmenite (Ilm) lamellae; (c) Opx megacryst with Cpx and Grt lamellae surrounded by coarser garnet crystals around the grain margin; (d) 'exsolution-like' Amp in Cpx megacryst; (e) mosaic textures with discrete Cpx, Grt, Opx and Amp; (f) Amp vein (dashed line) across the sample; (g) veined Amp reaction rim on Spl; (h, i) transition from lamellar to mosaic microstructure.

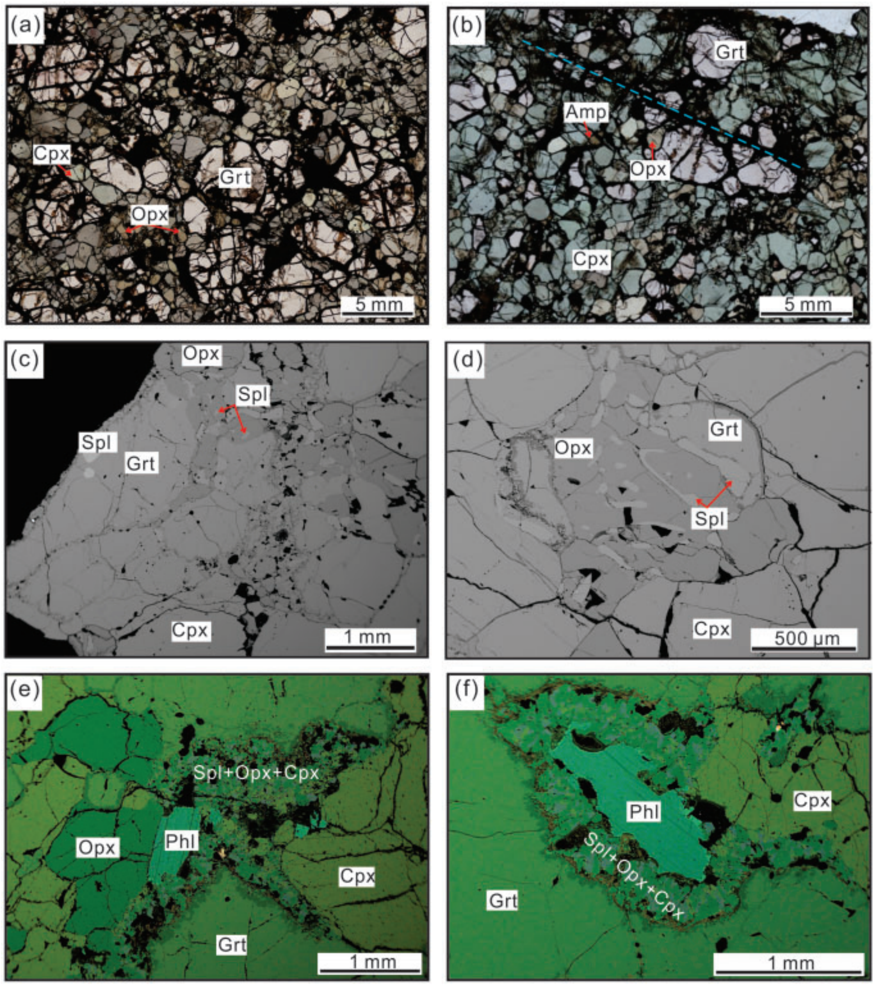

Fig. 4. Photomicrographs showing coarse textures and mineral assemblages. (a, b) Coarse Grt and relatively finer Cpx + Opx (plane-polarized light). Amp vein (dashed line) along the boundary of coarse Grt in (b) should be noted. (c, d) Spl–Grt relationship (BSE image); (c) spinel-core in direct contact with coarse garnet; (d) intergrowth of Opx and Spl. (e, f) Phlogopite in direct contact with discrete Cpx and Opx but not Grt in BM96-3 (false-colour SEM phase map).

garnet breakdown occurred following the reaction Grt + fluids → Cpx + Opx + Spl + Phl. Rare apatite (Ap) is also found in some samples.

The petrographic features described above suggest that the xenoliths from the Bullenmerri–Gnotuk maars preserve three distinct mineral assemblages. The first generation is composed of clinopyroxene megacrysts and small amounts of spinel and orthopyroxene. The second generation formed by the subsolidus re-equilibration of Cpx megacrysts through exsolution and recrystallization into dominantly Cpx, Grt, Opx and Amp. They were then intruded by veins of Amp and Phl, potentially related to an episode of volatile-rich metasomatism.

ANALYTICAL METHODS

All analyses were carried out at the Geochemical Analytical Unit (GAU) of the ARC Centre of Excellence for Core to Crust Fluid Systems (CCFS) and GEMOC, Macquarie University, Sydney, Australia.

Detailed analytical procedures for elemental X-ray mapping, and the analysis of major element compositions by electron probe microanalysis (EPMA) and of trace element compositions by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) have already been presented for the same set of xenoliths by Lu *et al.* (2017).

For Sr-, Nd-, and Hf-isotope analysis, about 200 mg of clean clinopyroxene and amphibole, and ~400 mg of clean garnet were separated from garnet websterites, acid-leached and digested, and processed using ion-exchange chromatographic column techniques described by Pin & Zalduegui (1997) and Blichert-Toft (1999). Sr- and Nd-isotopic ratios were analyzed by thermal ionization mass spectrometry (TIMS) using a Thermo Finnigan Triton system. Detailed analytical procedures have been described by Tilhac *et al.* (2017). Sr and Nd were loaded on single and double Re filaments, respectively. SRM 987 and JMC321 were measured to check instrument status and sensitivity, and yielded $^{87}Sr/^{86}Sr = 0{\cdot}710244 \pm 7$ (2σ) and $^{143}Nd/^{144}Nd = 0{\cdot}511129 \pm 2$ (2σ), respectively, which are within error of the GeoReM recommended values (http://georem.mpch-mainz.gwdg.de/). The USGS reference standard BHVO-2 yielded 0·703468 ± 8 (2σ) for $^{87}Sr/^{86}Sr$ and 0·512985 ± 3 (2σ) for $^{143}Nd/^{144}Nd$, also comparable with GeoReM recommended values of 0·703469 ± 34 and 0·512980 ± 12, respectively. Measured $^{87}Sr/^{86}Sr$ and $^{143}Nd/^{144}Nd$ ratios were normalized to $^{86}Sr/^{88}Sr$ = 0·1194 and $^{146}Nd/^{144}Nd$ = 0·7219 to correct for mass fractionation, respectively. Hf-isotope ratios were analysed using a Nu Plasma multi-collector (MC)-ICP-MS system. Standards JMC475 and BHVO-2 gave average $^{176}Hf/^{177}Hf$ ratios of 0·282156 ± 8 (2σ) and 0·283108 ± 24 (2σ), within error of the GeoReM recommended values of 0·282159 ± 26 and 0·283109 ± 12, respectively. Ratios were normalized to $^{179}Hf/^{177}Hf$ = 0·7325.

Sr isotopes in clinopyroxene and amphibole were also analyzed *in situ* using a 193 nm ArF excimer laser system attached to a Nu Plasma MC-ICP-MS system. Typical operating conditions for the laser were a 5 Hz repetition rate and an energy density of 7·5–9·3 J cm^{-2}, which provided the best signal stability and analytical precision. The spot size was 155 μm. Each analysis included 60 s of background followed by 140 s of data collection. The instrument was calibrated to the optimal conditions before analysis by measuring an SRM987 Sr solution. All analyses were done in time-resolved analysis mode, monitoring signals on each mass with the Nu Plasma analysis software to select the most stable interval of each run and avoid mixed ablation. Interference correlations were performed following Donnelly *et al.* (2012). The Batbjerg Cr-diopside (Batbjerg-1) reference material was run for monitoring instrument drift. In our sessions, analyses of Batbjerg-1 gave an average $^{87}Sr/^{86}Sr$ of 0·704454 ± 46 (2SD, n = 59), within the uncertainty of the published TIMS results ($^{87}Sr/^{86}Sr$ = 0·704474 ± 17; Neumann *et al.*, 2004).

RESULTS

Mineral compositions

Major element compositions

The major element compositions of minerals are listed in Table 2. Compositional profiles across grains of clinopyroxene and garnet indicate that the garnet pyroxenite xenoliths reached chemical equilibrium in terms of major elements (Lu *et al.*, 2017) and they show consistent compositions among different textures within individual samples (Table 2).

In the garnet pyroxenites studied here, Cpx is diopside ($En_{40–49}Fs_{1–8}Wo_{46–52}$) and has lower Mg# (79·3–89·5) and Cr_2O_3 (0·11–0·44 wt %) than the clinopyroxene of peridotite xenoliths from the same maars (Fig. 5a), but slightly higher Al_2O_3 (4·87–7·51 wt %) and similar CaO contents (Fig. 5b). Opx is enstatite ($En_{75–87}Fs_{12–23}$) and shows a positive correlation between MgO and Al_2O_3, but a negative correlation between MgO and CaO contents (Fig. 5c and d). Opx compositions are clearly correlated with those of coexisting Cpx. Grt is composed mainly of pyrope (55–71%), almandine (19–30%) and grossular (11–14%), with small amounts of spessartine (∼1%). It shows nearly constant CaO (5·26 wt % on average) contents with variable Cr_2O_3 contents (0·12–0·45 wt %) (Fig. 5e). Their Mg# and MgO contents show obvious positive correlations with those of coexisting Cpx. Spl is dominantly Mg–Al pleonaste and has lower Mg# at given Al_2O_3 content (or Cr#) than those from spinel pyroxenite xenoliths, which plot along the melt–peridotite reaction trend defined by composite xenoliths of spinel pyroxenite and spinel lherzolite from this area (Fig. 5f; Griffin *et al.*, 1984). Spinel also has slightly lower Cr_2O_3 but higher Ni contents at given Al_2O_3 than those from the spinel pyroxenites and peridotites. In addition, in sample BM96-2, spinel compositions show heterogeneity related to microstructure (Table 2): the Grt-rimming spinel has the highest Cr_2O_3 (∼14·2 wt %), whereas Amp-rimming spinel has intermediate Cr_2O_3 (∼5·55 wt %) and discrete spinel grains have the lowest Cr_2O_3 (∼3·74 wt %). These heterogeneous compositions are different from those produced by subsolidus reaction between spinel pyroxenite and peridotite (Fig. 5f). Sample BM96-2 also contains minor olivine (Ol), which has a much lower Mg# (∼80·4), but higher NiO (∼0·46 wt %), compared with olivine in peridotites (87·1–91·3 and 0·34–0·42 wt %, respectively). Ilmenite was found as exsolution lamellae in one Cpx megacryst; it has an MgO content of ∼7·0 wt %.

Amphiboles are all pargasites (Fig. 6a), but differ from the pargasitic amphiboles of peridotite xenoliths, which have higher Mg# and Cr_2O_3 contents (Fig. 6b). No correlation has been observed between their SiO_2 content and that of other oxides (e.g. Al_2O_3; Fig. 6c). The discrete Amp tends to have higher TiO_2 but lower K_2O than vein-forming Amp in samples BM96-2 and DR9748 (Fig. 6d). In sample BM96-3, Amp associated with Phl has the highest TiO_2 (∼3·97 wt %) and K_2O (∼1·81 wt %), and is essentially identical to the amphiboles in the late-stage wehrlite suite of xenoliths from the same locality (Griffin *et al.*, 1984). Phlogopite is found in one sample (BM96-3) and is inferred to be secondary, considering its microstructural position (Fig. 4e and f). It has homogeneous cores with moderate TiO_2 (∼6·50 wt %) and Mg# (∼79·1), whereas rims have higher TiO_2 (10·2–11·5 wt %) and lower Mg# (65·6–73·5), which represents a transformation to biotite. Rare apatite (Ap) in BME-1 has a F (0·98 wt %) content higher than Cl (0·29%) content.

Trace element compositions

The trace element compositions of minerals are listed in Tables 3–5. In most samples, no compositional gradients are observed in Grt and Cpx, except for sample DR10165, which shows obvious differences in heavy rare earth element (HREE) contents and normalized REE patterns related to their microstructural positions (Fig. 7).

Table 2: Representative major element compositions of minerals in the garnet pyroxenites from Lakes Bullenmerri and Gnotuk

Clinopyroxene

Sample:	BM 96-2	BM 99-6	BM 99-4		BME-1		BM 99-2		BM 96-3	BM 99-5	DR 9748*	GN35*		DR10165*	
Type:	M	M	H	M	H	M	H	M	M	H	M	H	M	H	M
SiO_2	51·2	51·3	51·0	51·4	51·5	51·8	50·9	51·2	51·4	51·3	50·9	51·1	51·4	51·1	51·4
TiO_2	0·37	0·36	0·73	0·79	0·43	0·42	0·40	0·40	1·02	0·80	0·88	0·68	0·62	0·53	0·51
Al_2O_3	6·62	6·75	7·14	7·51	6·33	6·42	6·43	6·34	6·20	4·87	5·98	7·29	7·11	6·5	6·02
Cr_2O_3	0·28	0·17	0·30	0·30	0·24	0·26	0·19	0·22	0·11	0·32	0·44	0·37	0·2	0·2	0·17
FeO	5·12	3·25	3·32	3·29	3·58	3·60	4·56	4·43	5·00	6·67	5·86	3·47	3·77	3·61	3·77
MnO	0·09	0·07	0·07	0·09	0·06	0·06	0·08	0·08	0·09	0·13	0·13	0·13	0·12		0·10
MgO	14·5	15·4	14·7	14·3	14·9	14·6	15·0	14·9	14·8	14·2	14·5	14·3	14·5	14·8	15·2
CaO	20·3	21·6	20·8	20·9	21·7	22·0	21·0	21·0	21·1	20·3	21·1	20·9	20·5	22·2	21·5
Na_2O	1·55	1·28	1·51	1·58	1·17	1·19	1·41	1·43	1·16	1·26	1·03	1·49	1·5	1·03	1·00
NiO	0·08	0·08	b.d.	b.d.	0·07	0·07	b.d.	b.d.	0·09	0·08	—	—	—	—	—
Total	100·1	100·2	99·6	100·2	99·9	100·5	99·9	100·1	100·8	99·9	100·8	99·7	99·8	99·9	99·7
Mg#	83·6	89·5	88·8	88·7	88·2	88·0	85·5	85·8	84·1	79·3	81·5	88·1	87·4	88·1	87·9

Orthopyroxene

Sample:	BM96-2	BM99-4			BME-1		BM99-2		BM96-3	BM99-5	DR9748*	GN35*		DR10165*	
Type:	M	E1	M	H	E1	M	E1	M	M	E1	M	E1	M	E1	M
SiO_2	53·3	54·5	54·8	55·0	55·2	54·4	53·5	53·4	54·9	53·3	53·6	54·2	54·3	54·2	54·4
TiO_2	0·08	0·11	0·10	0·12	0·07	0·08	0·06	0·05	0·22	0·25	0·15	0·12	0·10	—	0·10
Al_2O_3	4·63	4·61	4·72	4·79	4·61	4·60	4·70	4·60	4·12	3·69	4·07	4·99	4·92	4·84	4·77
Cr_2O_3	0·14	0·14	0·14	0·15	0·12	0·12	0·10	0·11	0·06	0·16	0·21	0·19	0·09	0·1	0·08
FeO	11·6	8·44	9·18	8·57	9·41	9·43	10·5	10·6	11·7	15·4	13·1	8·47	8·71	9·30	9·45
MnO	0·16	0·17	0·19	0·19	0·12	0·09	0·15	0·16	0·15	0·25	0·22	0·21	0·22	0·12	0·16
MgO	29·7	31·2	31·4	31·7	31·1	30·5	30·6	30·9	29·2	26·6	28·6	31·1	31·0	30·7	30·9
CaO	0·75	0·61	0·52	0·51	0·55	0·60	0·67	0·68	0·77	0·86	0·71	0·69	0·63	0·48	0·56
Na_2O	0·09	0·08	0·10	0·07	0·05	b.d.	0·09	0·08	0·08	0·11	0·08	—	0·08	—	0·06
NiO	0·12	b.d.	b.d.	b.d.	0·12	0·13	0·11	0·11	0·14	0·10	—	—	—	—	—
Total	100·6	100·0	101·1	101·1	101·4	100·0	100·5	100·5	101·4	100·7	100·8	100·0	100·0	99·7	100·5
Mg#	82·0	86·8	85·9	86·8	85·5	85·2	83·8	83·9	81·7	75·5	79·5	86·7	86·4	85·5	85·3

Garnet

Sample:	BM96-2	BM99-6	BM99-4			BME-1		BM99-2			BM96-3	BM99-5	DR9748*	GN35*		DR10165*
Type:	C	C	E1	E2	M	E1	C	E1	M	C	C	E1	M	E1	C	M
SiO_2	41·2	41·8	40·9	40·9	41·0	41·9	41·6	41·4	41·5	41·3	41·7	40·6	41·1	41·9	41·8	41·8
TiO_2	0·08	0·04	0·07	0·08	0·07	0·04	0·04	0·04	0·07	0·04	0·14	0·17	0·10	—	—	—
Al_2O_3	23·2	24·2	23·5	23·5	23·5	23·3	23·6	23·8	23·5	23·9	23·0	22·6	22·9	23·7	23·4	24·0
Cr_2O_3	0·45	0·18	0·24	0·25	0·25	0·27	0·24	0·21	0·23	0·26	0·12	0·44	0·64	0·32	0·52	0·17
FeO	13·3	9·3	10·7	10·5	10·8	10·9	11·7	12·3	12·6	11·9	13·7	16·6	15·0	10·1	10·6	11·6
MnO	0·41	0·31	0·50	0·51	0·52	0·25	0·30	0·42	0·42	0·42	0·42	0·57	0·56	0·54	0·56	0·44
MgO	17·4	19·6	19·1	19·3	19·0	17·8	18·6	17·8	18·3	17·7	15·9	14·6	15·5	18·6	18·4	17·7
CaO	5·30	5·47	4·28	4·20	4·24	5·80	5·42	5·46	5·39	5·46	5·61	5·79	5·81	5·23	5·16	5·53
Na_2O	b.d.	b.d.	b.d.	b.d.	b.d.	0·24	b.d.	b.d.	b.d.	b.d.	b.d.	0·02	—	—	—	—
Total	101·3	101·0	99·3	99·2	99·5	100·4	101·5	101·3	102·0	101·0	100·5	101·4	101·6	100·4	100·4	101·3
Mg#	70·2	79·1	76·2	76·7	76·0	74·7	74·1	72·3	72·4	72·9	67·6	61·4	64·7	76·6	75·6	73·1

(continued)

Table 2: Continued

	Amphibole												Phlogopite			Ap
Sample:	BM96-2		BM99-6	BM99-4		BME-1	BM96-3	BM99-5	DR9748*		GN35	DR10165*	BM96-3			BME-1
Type:	M	vein	M	M	vein	E1	M	E1	M	vein	M	M	Core	Rim1	Rim2	M
SiO_2	42·2	41·8	42·1	41·9	42·7	42·0	41·9	40·9	42·0	42·1	42·9	42·5	36·3	35·3	33·5	
TiO_2	1·57	0·92	1·47	2·56	0·84	1·63	3·97	2·24	3·47	1·62	2·60	2·09	6·50	10·2	11·5	
Al_2O_3	15·5	15·9	16·2	15·6	15·8	15·7	13·4	14·2	14·8	15·6	16·0	15·9	15·5	15·6	15·7	
Cr_2O_3	0·29	0·26	0·24	0·43	0·35	0·35	0·13	0·42	0·54	0·40	0·34	0·29	0·14	0·20	—	
FeO	6·86	7·14	4·65	5·03	5·34	5·33	6·97	9·72	8·54	8·46	5·92	5·38	7·90	9·01	10·8	
MnO	0·06	0·06	0·04	0·07	0·10	0·05	0·05	0·10	—	0·09	0·10	—	0·04	0·06	0·08	
MgO	16·0	15·9	17·1	16·7	16·9	16·7	14·5	14·0	14·9	14·7	16·8	16·8	16·6	14·0	11·5	
CaO	10·6	10·5	11·0	11·0	11·0	11·0	11·4	10·4	10·7	10·6	10·5	11·1	0·05	0·05	0·08	52·5
Na_2O	3·28	3·01	3·36	3·93	3·44	3·58	2·50	3·21	3·35	2·80	3·69	3·60	0·51	0·42	0·30	
K_2O	0·97	1·56	0·57	0·07	0·90	0·10	1·81	1·03	0·30	1·72	0·36	0·14	8·45	8·20	7·64	
P_2O_5																40·8
NiO	0·12	0·14	0·14	b.d.	b.d.	0·14	0·16	b.d.	0·14	0·11	0·11	—	0·30	0·22	—	
Cl	0·07	0·09	0·07	0·04	0·04	b.d.	0·18	0·04	—	—	—	—	0·27	0·10	0·08	0·29
F	b.d.	b.d.	b.d.	b.d.	b.d.	0·13	b.d.	0·07	—	—	—	—	0·07	0·12	0·12	0·98
Total	97·4	97·3	96·9	97·3	97·3	96·6	97·0	96·4	98·8	98·2	99·3	97·8	92·5	93·5	91·3	94·6
Mg#	80·6	79·9	86·7	85·6	84·9	84·8	78·7	72·0	75·7	75·6	83·5	84·8	78·9	73·5	65·6	

	Spinel								Olivine	Ilmenite
Sample:	BM96-2			BM99-6	BME-1	DR10165*		GN35*	BM96-2	BM99-5
Type:	M	M†	M‡	M	M†	M†	M	M	M	E1
SiO_2	0·06	0·07	0·07	0·09	0·04	—	—	—	39·5	0·0
TiO_2	0·10	0·34	0·08	0·05	0·05	—	—	0·12	b.d.	47·1
Al_2O_3	59·7	47·2	57·7	64·6	61·3	64·2	64·4	64·6	0·03	0·35
Cr_2O_3	3·74	14·24	5·55	1·94	4·23	2·84	2·31	3·49	b.d.	0·51
FeO	17·8	22·0	18·3	11·3	13·8	13·7	13·7	13·0	18·0	39·4
MnO	0·08	0·11	0·10	0·07	0·06	b.d.	0·05	0·14	0·15	0·35
MgO	17·3	14·7	16·7	20·5	19·1	19·5	19·6	19·6	41·6	7·0
CaO	b.d.	0·02	b.d.	0·05	b.d.	—	—	—	0·07	0·08
ZnO	0·23	0·18	0·24	0·13	0·36	—	—	—	—	—
NiO	0·59	0·42	0·51	0·65	0·65	0·27	0·24	0·33	0·48	0·15
Total	99·6	99·5	99·3	99·3	99·6	100·5	100·3	101·2	99·9	95·3
Mg#	63·3	54·4	62·0	76·3	71·2	71·7	71·9	72·9	80·4	24·0
Cr#	4·04	16·83	6·06	1·97	4·42	2·88	2·35	3·50		

M, minerals in the mosaic area; H, host minerals with exsolution textures; E1, minerals as exsolved lamellae in Cpx megacryst; E2, mineral as exsolved lamellae in Opx megacryst; C, Grt in the coarse area; b.d., below detection limit; —, no value.
*Data from Griffin *et al.* (1984) and O'Reilly & Griffin (1985).
†Grt-rimmed spinel.
‡Amp-rimmed spinel.

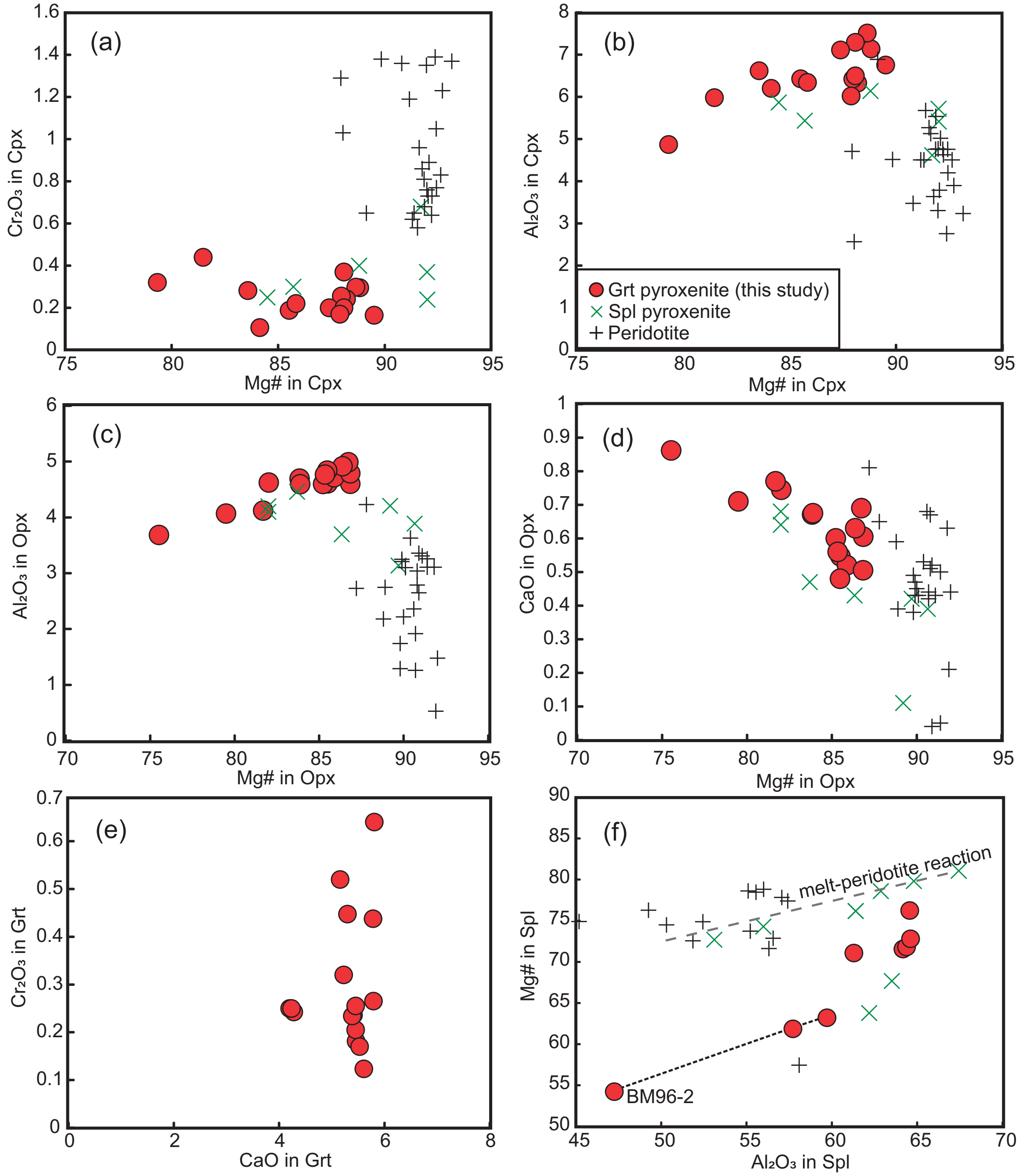

Fig. 5. Compositional variations of Cpx (a, b), Opx (c, d), Grt (e), and Spl (f) in garnet pyroxenites. Spinel-pyroxenitic and peridotitic minerals from this locality are shown for comparison (Griffin *et al.*, 1984; Powell, 2005). Dashed line in (f) represents melt–peridotite reaction observed in composite xenoliths of spinel pyroxenite and lherzolite (GN23; Griffin *et al.*, 1984); dotted line links Spl of sample BM96-2.

Cpx exhibits flat to variably light REE (LREE)-enriched patterns ($[La/Nd]_{CN} = 0{\cdot}77$–$2{\cdot}22$; Fig. 7a; Table 3) and fractionated middle REE (MREE)–HREE patterns ($[Sm/Yb]_{CN} = 4{\cdot}82$–$60{\cdot}0$, but mostly less than 19·2) owing to the exsolution of Grt from Cpx megacrysts. Most clinopyroxenes have slightly positive Eu anomalies ($Eu/Eu^* = 1{\cdot}04$–$1{\cdot}38$), positive Sr anomalies and negative Nb–Ta anomalies, but no obvious Ti anomaly (Fig. 7b). Zr is fractionated from Hf ($[Zr/Hf]_{PM} = 0{\cdot}46$–$0{\cdot}74$). In contrast, Cpx in samples DR10165 and BME-1 shows higher LREE contents, steeper negative MREE–HREE slopes ($[Sm/Yb]_{CN} = 29{\cdot}6$–$60{\cdot}0$) and marked negative Sr and Pb anomalies along with strongly negative high field strength element (HFSE) anomalies (Fig. 7a and b), but unfractionated Zr/Hf ($[Zr/Hf]_{PM} = 0{\cdot}90$–$1{\cdot}11$).

Amp displays LREE-enriched and HREE-depleted REE patterns ($[La/Yb]_{CN} = 4{\cdot}50$–$90{\cdot}6$), similar to the pattern of coexisting Cpx, but at slightly higher contents (Fig. 7c; Table 4). Discrete Amp has highly fractionated Nb/Ta ($[Nb/Ta]_{PM} = 1{\cdot}39$–$5{\cdot}63$) and displays pronounced

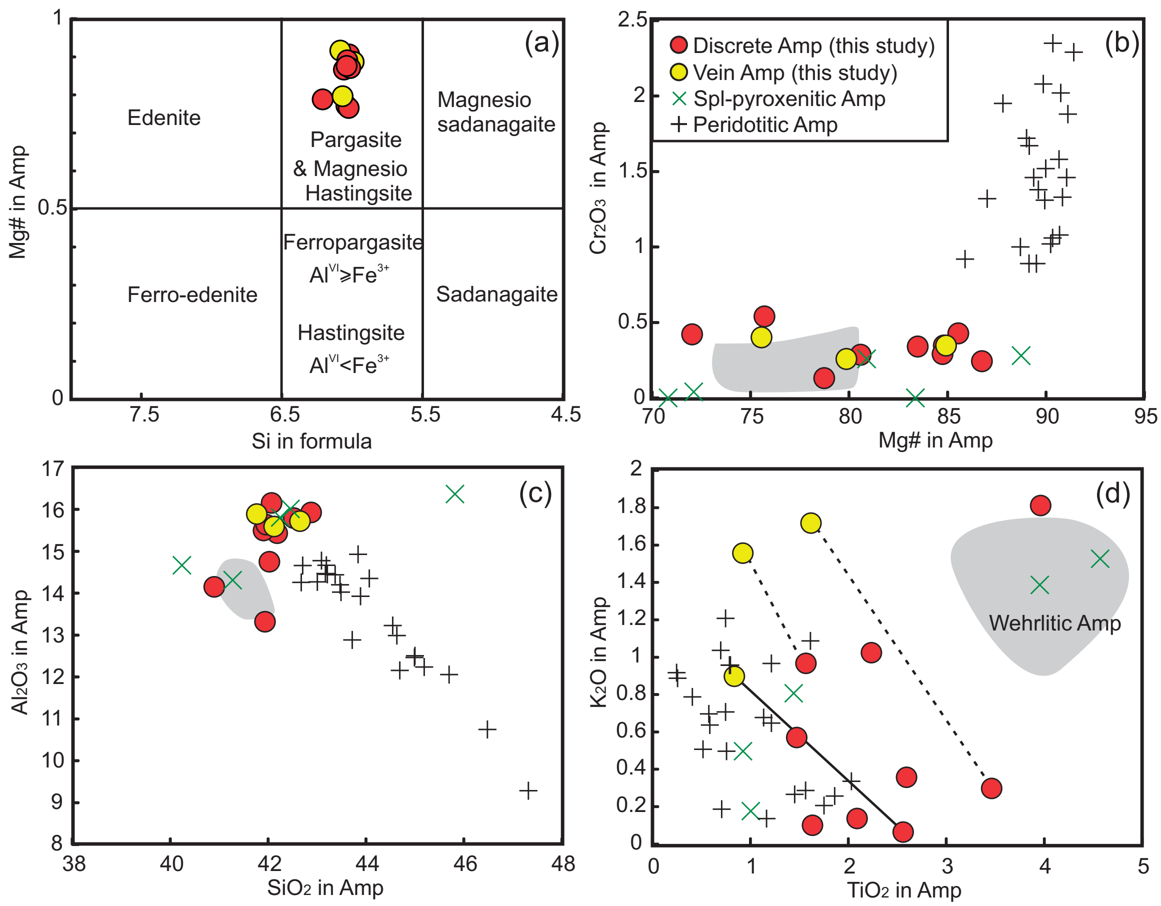

Fig. 6. Compositional variations of Amp in the garnet pyroxenite xenoliths from Lakes Bullenmerri and Gnotuk. Amp from spinel pyroxenite and peridotite in this locality are shown for comparison (Griffin *et al.*, 1984; Powell, 2005). Grey shaded areas represent Amp from wehrlite xenoliths in this area (Griffin *et al.*, 1984).

positive anomalies in Pb, Sr and Ti (Fig. 7d) and variable negative anomalies in Zr and Hf. In contrast, vein-related amphiboles in samples DR9748 and BM96-2 have slightly higher incompatible-element concentrations than the discrete ones, and they show pronounced positive HFSE anomalies (Fig. 7d) with little Nb–Ta fractionation ($[Nb/Ta]_{PM} = 1{\cdot}07$–$1{\cdot}51$).

Garnet displays LREE-depleted REE patterns ($[Sm/La]_{CN} = 15{\cdot}34$–$131$) with near-flat ($[Gd/Yb]_{CN} = 0{\cdot}92$; e.g. BM99-6) to elevated ($[Gd/Yb]_{CN} = 0{\cdot}24$; e.g. BM99-4) MREE–HREE slopes (Fig. 7e; Table 5). All garnets also show negative Ti anomalies but variable Ti contents (Fig. 7f).

Bulk-rock reconstructions

Reconstruction of the bulk compositions of the garnet websterites used the analysed compositions of Cpx, Grt, Opx and Amp and their modal proportions as determined by point counting (Supplementary Data Electronic Appendix 1; supplementary data are available for downloading at http://www.petrology.oxfordjournals.org). Because compositional heterogeneities have been observed in only a few grains, we assumed that most minerals have reached equilibrium.

Point-counted modes slightly differ from mineral modes calculated using the least-squares methods modified from MINSQ (Table 1; Herrmann & Berry, 2002), but both methods yielded results that are within 5% relative for most major elements and 20% for compatible and moderately incompatible trace elements (i.e. DR9748; Supplementary Data Electronic Appendix 1) when compared with measured whole-rock compositions (Griffin *et al.*, 1988). Calculated bulk-rock compositions are also comparable with those of primary clinopyroxene before subsolidus re-equilibration (Lu *et al.*, 2017), indicating that Cpx was the main liquidus phase (Griffin *et al.*, 1984; Lu *et al.*, 2017).

The Grt websterites have MgO of 17·1–20·8 wt %, CaO of 8·40–15·2 wt %, SiO_2 of 45·3–50·7 wt %, Al_2O_3 of 5·73–13·9 wt %, FeO of 5·63–11·1 wt %, Na_2O of 0·28–1·08 wt % and TiO_2 of 0·23–0·88 wt % (Fig. 8). These major element compositions are similar to those of high-MgO Grt-pyroxenite cumulates from the Sierra Nevada (Lee *et al.*, 2006). Both suites plot within the field of experimentally produced pyroxenite cumulates from high-*P* hydrous basaltic andesites (Müntener *et al.*, 2001; Fig. 8a and c). In the pseudoternary forsterite–Ca-Tschermak pyroxene–quartz (Fo–CaTs–Qz) system projected from diopside (Di), the Grt websterites plot along

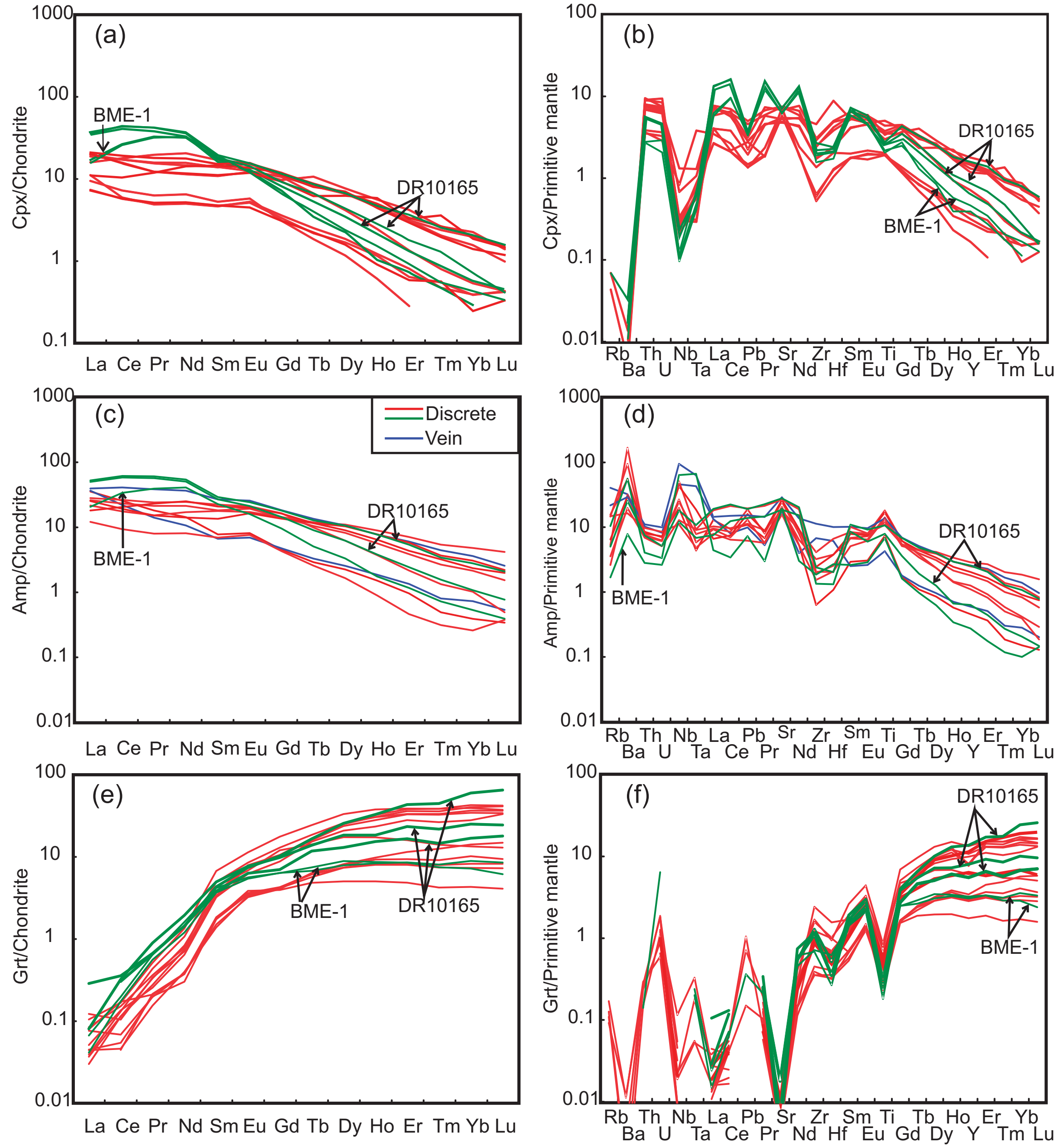

Fig. 7. Chondrite-normalized REE (a, c and e) and primitive-mantle normalized trace element (b, d and f) patterns of Cpx (a, b), Amp (c, d) and Grt (e, f) in garnet pyroxenite xenoliths from Lakes Bullenmerri and Gnotuk. Chondrite and primitive-mantle compositions from McDonough & Sun (1995). The green lines are minerals from samples BME-1 and DR10165.

the En–CaTs join, as do pyroxenite cumulates from other localities worldwide (Fig. 9). In contrast, the spinel pyroxenites from the same locality (Griffin *et al.*, 1988), characterized by high MgO and Cr_2O_3 and low CaO (Fig. 8), plot between the Fo–An join and the En–CaTs join (Fig. 9), suggesting they are the products of melt–peridotite reaction (Garrido & Bodinier, 1999; Rapp *et al.*, 1999; Liu *et al.*, 2005; Bodinier *et al.*, 2008; Lambart *et al.*, 2012; Tilhac *et al.*, 2016).

The Grt websterites show a limited range of low Ni contents (mostly <500 ppm) compared with the associated peridotite xenoliths (1279–2111 ppm), suggesting that the abundance of olivine in the original cumulates was low. They also have slightly lower Cr (614–2273 ppm) contents than spinel websterite and peridotite xenoliths (Griffin *et al.*, 1988; Powell, 2005; 2810–5436 ppm and 1680–3644 ppm, respectively).

The garnet websterites have relatively flat REE patterns ($[La/Yb]_{CN} = 1{\cdot}15$–$3{\cdot}41$) with small positive Eu anomalies ($Eu/Eu^* = 0{\cdot}90$–$1{\cdot}37$), although they display variable ΣREE (8·62–37·4) (Fig. 10). The flat REE patterns are similar to those of peridotite xenoliths metasomatized by silicate melts from the same locality (Powell *et al.*, 2004). Sample BM96-3 shows HREE enrichment relative to the flat LREE ($[Sm/Yb]_{CN} = 0{\cdot}71$; $[La/Nd]_{CN} = 1{\cdot}14$), potentially indicating a slight overestimation of the modal

Table 3: Trace element compositions (ppm) of clinopyroxene in the garnet pyroxenites from Lakes Bullenmerri and Gnotuk

Sample:	BM96-2	BM99-6	BM99-4		BME-1		BM99-2		BM96-3	BM99-5	DR9748	GN35		DR10165		
Type:	M	M	H	M	H	M	H	M	M	H	M	H	M	H	M1	M2
Li	3·24	1·57	4·62	4·04	2·15	2·58	2·13	2·12	3·09	4·07	3·84	2·21	2·04	—	2·76	2·81
Be	0·17	0·13	0·25	0·28	0·29	0·29	0·11	0·09	0·23	0·48	0·32	b.d.	b.d.	—	b.d.	b.d.
B	1·87	2·10	10·06	8·31	2·46	1·53	1·73	2·42	1·74	1·74	0·76	9·80	8·60	—	10·09	10·51
Sc	28·4	17·2	37·0	39·6	22·0	30·7	26·9	34·0	34·3	54·6	49·1	43·0	50·5	29·8	31·6	39·2
Ti	2285	2450	4066	4259	2478	2632	2302	2337	6059	4539	5280	4212	3597	3127	3077	3240
V	177	187	351	346	230	241	244	254	487	392	358	277	263	198	200	191
Cr	1816	1173	1794	1924	1407	1601	1245	1455	597	2047	2450	2397	1479	1334	1226	1233
Co	41·0	27·7	25·6	23·9	34·6	36·1	37·6	37·0	43·8	45·8	46·6	23·9	25·6	24·6	23·5	23·2
Ni	472	439	227	227	428	444	410	387	507	385	511	304	314	294	287	259
Cu	1·00	1·82	1·44	1·26	0·75	0·75	1·49	1·41	1·82	1·49	3·47	1·98	1·98	1·05	1·03	1·01
Zn	21·1	6·4	14·5	11·8	18·7	18·6	24·3	23·8	33·7	52·3	47·2	6·6	6·4	6·1	6·3	6·4
Ga	8·74	9·78	11·08	10·59	9·73	10·44	12·45	12·43	16·80	11·47	12·11	5·26	4·09	4·33	3·76	2·94
Rb	b.d.	b.d.	0·04	b.d.	b.d.	b.d.	0·03	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.	0·04
Sr	146	137	149	147	126	125	114	115	104	109	135	95	97	128	138	138
Y	1·56	0·72	5·07	5·62	1·68	1·99	1·21	1·36	3·06	6·54	6·23	6·28	7·68	2·82	3·79	6·70
Zr	6·64	14·27	25·35	26·01	16·44	19·90	5·49	6·23	47·95	30·48	32·16	18·51	18·07	21·85	22·78	27·70
Nb	0·14	0·55	0·20	0·17	0·08	0·06	0·18	0·22	0·86	0·07	0·45	0·20	0·22	0·08	0·12	0·15
Ba	0·05	0·12	0·09	0·05	0·09	0·07	0·04	0·05	0·12	0·06	0·07	0·09	0·11	0·11	0·12	0·21
La	2·64	2·24	4·98	4·66	4·03	3·72	1·71	1·74	4·45	2·61	3·77	4·80	4·80	8·07	8·47	8·76
Ce	3·64	4·43	11·73	10·50	16·17	15·82	3·53	3·51	10·54	6·38	10·95	9·49	9·66	24·72	27·28	26·93
Pr	0·46	0·60	1·72	1·50	3·09	3·03	0·48	0·48	1·47	1·13	1·87	1·17	1·16	3·62	4·00	4·06
Nd	2·34	3·03	8·29	7·33	14·89	15·23	2·42	2·41	6·82	6·69	9·59	5·46	5·32	15·28	16·80	17·29
Sm	0·71	0·81	2·32	2·12	2·68	2·97	0·72	0·72	2·15	2·23	2·60	1·69	1·66	2·49	2·78	2·99
Eu	0·31	0·33	0·91	0·89	0·71	0·69	0·26	0·30	0·73	0·79	0·89	0·70	0·71	0·74	0·83	0·90
Gd	0·74	0·64	2·31	2·20	1·49	1·35	0·64	0·68	2·17	2·43	2·44	1·91	1·99	1·66	1·90	2·36
Tb	0·09	0·07	0·30	0·31	0·13	0·15	0·08	0·08	0·25	0·33	0·31	0·23	0·40	b.d.	0·25	0·32
Dy	0·47	0·30	1·78	1·67	0·57	0·61	0·40	0·43	1·06	1·73	1·72	1·58	1·94	0·88	1·10	1·71
Ho	0·07	0·03	0·25	0·26	0·06	0·09	0·05	0·07	0·14	0·28	0·26	0·33	0·32	b.d.	0·16	0·27
Er	0·13	0·05	0·50	0·52	0·12	0·15	0·10	0·11	0·22	0·56	0·52	0·54	0·69	0·22	0·30	0·61
Tm	0·01	b.d.	0·05	0·05	0·01	0·01	0·01	0·01	0·02	0·07	0·06	0·09	0·07	b.d.	0·03	0·07
Yb	0·07	b.d.	0·24	0·27	0·05	0·07	0·04	0·07	0·09	0·35	0·34	0·32	0·38	0·10	0·12	0·35
Lu	0·01	0·01	0·03	0·03	b.d.	0·01	0·01	0·01	0·01	0·04	0·04	0·04	0·04	0·01	0·01	0·04
Hf	0·35	0·58	1·18	1·08	0·49	0·59	0·29	0·36	2·50	1·38	1·41	0·67	0·68	0·61	0·63	0·67
Ta	0·02	0·14	0·02	0·02	0·03	0·02	0·03	0·03	0·05	0·02	0·04	0·01	0·01	0·02	0·02	0·02
Pb	0·42	0·20	0·75	0·69	0·50	0·52	0·22	0·20	0·34	0·29	0·47	0·43	0·42	0·54	0·55	0·54
Th	0·55	0·29	0·71	0·72	0·22	0·22	0·31	0·28	0·56	0·63	0·76	0·61	0·60	0·46	0·43	0·46
U	0·13	0·06	0·17	0·19	0·06	0·04	0·07	0·07	0·13	0·15	0·15	0·14	0·14	0·10	0·09	0·09

M, Cpx in the mosaic area; H, host Cpx with exsolution textures; b.d., below detection limit; —, no value.

Table 4: Trace element compositions (ppm) of amphibole in the garnet pyroxenites from Lakes Bullenmerri and Gnotuk

Sample:	BM96-2		BM99-6	BM99-4	BME-1	BM96-3	BM99-5	DR9748		GN35	DR10165	
Type:	M	vein	M	M	E1	M	E1	M	vein	M	M1	M2
Li	1·97	1·97	1·09	2·91	2·06	0·95	3·14	2·29	3·26	0·97	1·59	1·63
Be	0·42	0·53	0·24	0·58	0·58	0·33	0·85	0·49	0·70	0·30	0·30	0·27
B	6·52	6·94	6·4	6·00	7·26	5·41	7·85	0·69	1·34	2·76	2·97	2·85
Sc	19·0	20·3	13·0	28·5	16·6	26	45·8	38·2	37	41·5	28·3	38·4
Ti	8856	5149	8224	14344	9551	21921	17059	20413	16530	16103	13087	12879
V	265	180	248	476	336	649	468	494	386	322	294	269
Cr	1829	1665	1547	2644	2237	850	2866	4553	3045	2302	1944	1510
Co	79·7	77·6	53·6	52·0	67·4	79·4	75·5	79·1	81·0	47·5	46·6	47·2
Ni	1212	1053	1040	613	1072	1274	782	1171	1032	773	723	704
Cu	1·71	1·81	3·31	2·00	1·55	2·04	2·08	3·91	4·56	0·88	0·81	0·75
Zn	45·0	42·9	12·8	22·8	35·1	54·7	91·3	70·9	81·7	9·70	9·10	9·30
Ga	36·1	27·1	36·4	27·4	18·5	73	23·6	27·9	30·8	49·5	19·1	20·9
Rb	9·35	24·4	6·25	1·57	1·01	3·17	8·88	3·88	13·1	8·77	2·14	3·02
Sr	409	327	386	493	371	359	384	525	577	311	530	556
Y	1·96	2·56	1·18	7·75	2·72	4·76	12·36	8·77	11·07	10·71	5·42	11·1
Zr	6·64	71·3	19·6	20·7	14·1	43·3	33·9	25·7	119	16·1	21·6	25·3
Nb	13·0	30·2	41·9	16·3	7·99	33·5	7·27	28·4	62·5	17·6	8·48	11·9
Ba	365	213	371	211	52·0	1109	109	198	193	623	130	168
La	8·4	8·68	2·88	6·17	4·91	6·68	5·13	4·30	9·38	6·00	11·8	12·4
Ce	15·7	13·5	5·7	14·2	20·7	16·2	10·8	12·5	25·1	11·8	35·7	37·7
Pr	1·73	1·34	0·76	2·03	3·73	2·29	1·45	2·2	3·66	1·44	5·43	5·76
Nd	6·68	4·99	3·78	9·97	19·1	11·5	8·00	11·7	17·1	6·60	23·6	25·4
Sm	1·19	1·02	1·06	2·78	3·48	3·42	2·73	3·2	4·12	2·07	4·11	4·46
Eu	0·47	0·4	0·44	1·12	0·93	1·22	1·00	1·16	1·49	0·95	1·23	1·41
Gd	0·93	0·97	0·88	2·94	1·97	3·38	2·97	3·34	3·76	2·79	2·97	3·72
Tb	0·11	0·12	0·1	0·39	0·19	0·37	0·44	0·43	0·51	0·43	0·34	0·48
Dy	0·57	0·65	0·42	2·09	0·84	1·58	2·81	2·29	2·72	2·63	1·58	2·70
Ho	0·09	0·11	0·05	0·32	0·10	0·21	0·51	0·37	0·44	0·44	0·22	0·44
Er	0·16	0·22	0·08	0·64	0·19	0·36	1·18	0·72	1·00	0·96	0·40	0·90
Tm	0·01	0·02	0·01	0·07	0·02	0·03	0·14	0·08	0·11	0·09	0·04	0·09
Yb	0·07	0·12	0·04	0·37	0·09	0·17	0·82	0·4	0·61	0·52	0·18	0·47
Lu	0·01	0·01	0·01	0·04	0·01	0·01	0·11	0·05	0·07	0·06	0·02	0·05
Hf	0·31	1·74	0·76	0·72	0·37	1·87	1·07	1·06	2·85	0·59	0·58	0·57
Ta	0·16	1·59	2·47	0·29	0·25	0·34	0·18	0·69	2·34	0·22	0·34	0·40
Pb	1·64	1·49	1·09	2·84	2·09	1·7	0·93	2·4	2·29	1·97	2·93	2·86
Th	0·64	0·82	0·32	0·57	0·22	0·61	0·78	0·68	0·88	0·59	0·55	0·59
U	0·15	0·17	0·07	0·12	0·05	0·13	0·17	0·14	0·20	0·13	0·10	0·10

M, Amp in the mosaic area; E1, Amp as exsolution-like lamellae in Cpx megacryst.

Table 5: Trace element compositions (ppm) of garnet in the garnet pyroxenites from Lakes Bullenmerri and Gnotuk

Sample:	BM96-2	BM99-6	BM99-4			BME-1		BM99-2			BM96-3	BM99-5	DR9748	GN35		DR10165		
Type:	C	C	E1	E2	M	E1	C	E1	M	C	C	E1	M	E1	C	E1	M	M
Li	0·31	b.d.	0·65	0·45	0·28	b.d.	0·33	b.d.	b.d.	0·16	0·38	0·52	0·48	b.d.	b.d.	b.d.	b.d.	b.d.
B	1·49	2·21	12·2	16·9	12·8	1·79	1·75	1·95	2·25	2·93	1·18	1·68	0·95	b.d.	10·8	b.d.	9·78	10·9
Sc	88·0	63·7	89·3	82·8	84·4	58·3	53·3	90·2	83·9	113	87·9	153	110	136	131	78·1	97·1	100
Ti	557	285	368	354	321	222	271	381	423	359	889	1043	978	569	606	356	381	333
V	51·6	38·8	59·8	58·6	55·6	47·7	57·2	62·5	66·0	62·5	133	116	118	67·3	61·4	58·5	44·6	54·1
Cr	2943	1187	1240	1283	1302	1622	1650	1415	1431	1683	747	2908	3191	2094	2091	1538	1312	1519
Co	75·4	58·4	43·3	46·4	42·9	75·1	74·3	69·8	71·2	72·4	75·4	69·3	73·3	47·9	49·6	51·0	51·2	46·7
Ni	48·4	46·7	21·8	22·7	20·6	44·0	40·5	45·9	45·1	38·7	56·6	37·2	54·9	37·2	37·8	30·4	29·4	25·7
Cu	0·17	1·27	0·69	b.d.	0·03	0·27	0·14	b.d.	0·17	0·12	0·15	0·30	0·17	0·13	0·09	0·41	0·12	b.d.
Zn	17·77	6·56	14·24	18·05	14·90	18·56	19·12	23·71	24·11	22·19	29·96	41·46	38·60	7·46	6·94	6·14	6·94	6·99
Ga	6·45	4·20	3·78	3·93	3·95	4·63	4·98	6·48	6·90	6·16	8·58	7·10	7·10	2·61	3·46	2·85	1·58	2·36
Rb	0·07	b.d.	0·10	0·01	b.d.	b.d.	b.d.	b.d.	b.d.	0·08	b.d.	0·06	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.
Sr	0·13	0·13	0·20	0·07	0·08	0·07	0·12	0·10	0·09	0·09	0·14	0·17	0·14	0·07	0·07	0·37	0·10	0·15
Y	12·88	7·66	35·62	33·27	35·37	14·11	13·38	13·83	12·42	14·97	26·33	57·97	46·71	49·56	43·30	24·08	34·86	59·57
Zr	4·82	13·10	9·46	9·01	10·04	7·30	7·65	3·78	3·80	4·16	25·82	18·04	16·93	9·21	9·85	11·84	12·38	13·77
Nb	0·01	0·08	0·01	0·01	0·01	b.d.	b.d.	0·02	0·01	0·02	0·04	0·01	0·03	0·02	0·01	b.d.	0·01	0·01
Ba	0·01	0·22	0·08	b.d.	0·00	0·02	b.d.	0·00	0·01	0·01	0·03	0·08	b.d.	b.d.	b.d.	b.d.	0·09	0·07
La	0·02	0·02	0·03	0·01	0·01	0·02	0·01	b.d.	0·01	0·01	0·02	0·03	0·01	0·02	0·01	b.d.	0·02	0·07
Ce	0·04	0·11	0·06	0·07	0·07	0·12	0·11	0·03	0·03	0·03	0·09	0·10	0·08	0·08	0·08	0·19	0·20	0·22
Pr	0·02	0·02	0·03	0·03	0·03	0·06	0·05	0·02	0·01	0·02	0·04	0·05	0·04	0·02	0·02	0·06	0·09	0·07
Nd	0·18	0·18	0·34	0·40	0·36	0·78	0·58	0·18	0·19	0·14	0·38	0·65	0·52	0·30	0·31	0·70	0·94	0·69
Sm	0·29	0·28	0·52	0·58	0·54	0·67	0·59	0·24	0·22	0·28	0·70	1·08	0·79	0·43	0·52	0·61	0·79	0·69
Eu	0·22	0·23	0·46	0·48	0·47	0·34	0·33	0·20	0·20	0·21	0·42	0·69	0·52	0·36	0·38	0·38	0·47	0·45
Gd	0·92	0·85	2·21	2·27	2·28	1·37	1·37	0·87	0·89	0·91	2·60	3·82	2·80	1·89	1·90	1·51	2·12	2·21
Tb	0·24	0·19	0·63	0·66	0·65	0·29	0·27	0·25	0·22	0·25	0·56	0·96	0·73	b.d.	0·60	0·46	0·55	0·65
Dy	1·98	1·33	6·39	6·74	6·74	2·37	2·10	2·11	1·96	2·17	4·65	8·94	6·94	6·17	5·60	3·48	4·92	6·88
Ho	0·49	0·30	1·68	1·95	1·84	0·51	0·50	0·55	0·48	0·58	1·03	2·27	1·83	b.d.	1·40	0·92	1·10	1·97
Er	1·45	0·84	5·78	6·69	6·34	1·48	1·49	1·63	1·39	2·00	2·79	6·82	5·84	6·73	4·92	2·91	4·10	7·63
Tm	0·21	0·11	0·88	1·04	0·97	0·20	0·22	0·24	0·20	0·34	0·38	1·04	0·90	b.d.	0·71	0·39	0·59	1·21
Yb	1·47	0·77	6·20	7·70	7·29	1·29	1·59	1·82	1·31	2·54	2·40	7·09	6·60	8·47	5·02	3·02	4·50	10·86
Lu	0·22	0·11	0·92	1·14	1·11	0·16	0·23	0·25	0·19	0·40	0·35	1·01	0·98	1·35	0·89	0·48	0·66	1·76
Hf	0·12	0·29	0·19	0·19	0·17	0·08	0·18	0·10	0·09	0·09	0·45	0·27	0·29	0·14	0·20	0·10	0·10	0·15
Ta	0·01	0·01	b.d.	b.d.	b.d.	0·01	0·01	b.d.	b.d.	0·00	b.d.	0·00	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.
Pb	b.d.	0·16	b.d.	b.d.	0·02	b.d.	0·06	b.d.	b.d.	b.d.	b.d.	0·11	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.
Th	0·02	b.d.	0·02	b.d.	0·01	b.d.	0·01	b.d.	b.d.	b.d.	0·02	0·02	b.d.	b.d.	b.d.	b.d.	b.d.	b.d.
U	0·02	0·02	0·01	0·04	0·02	b.d.	0·13	0·01	b.d.	b.d.	0·04	0·04	0·02	b.d.	0·03	b.d.	b.d.	b.d.

C, Grt in the coarse area; M, Grt in the mosaic area; E1, Grt as exsolution lamellae in Cpx megacryst; E2, Grt as exsolution lamellae in Opx megacryst; b.d., below detection limit.

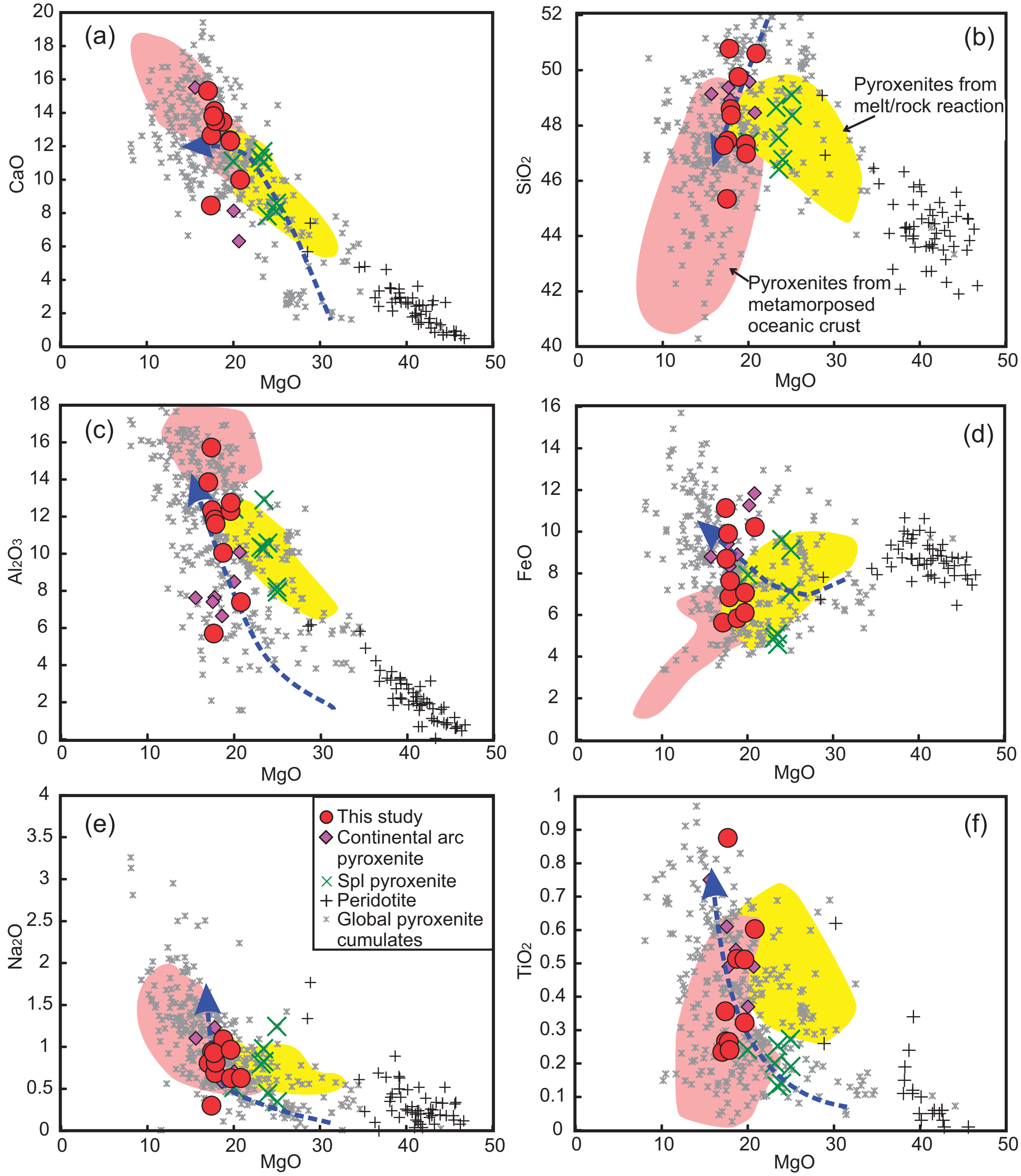

Fig. 8. Whole-rock major oxides vs MgO for garnet pyroxenites (red circle; this study), spinel pyroxenites (Griffin *et al.*, 1988) and peridotite xenoliths (O'Reilly & Griffin, 1988; Stolz & Davies, 1988; Powell, 2005) from Lakes Bullenmerri and Gnotuk. High-MgO Grt-pyroxenite cumulates (pink diamond) are from the Sierra Nevada continental arc (Lee *et al.*, 2006); the worldwide ('Global') database of pyroxenite cumulates, pyroxenites from metamorphosed oceanic crust (pink shaded area) and from melt–rock reaction (yellow shaded area) are from Xiong *et al.* (2014). Dashed line (blue) represents fractional crystallization trend defined by experimentally derived pyroxenite cumulates from hydrous basaltic andesite at 1·2 GPa (Müntener *et al.*, 2001).

proportion of garnet. Samples BME-1 and DR10165 have convex-upward REE patterns with elevated downward-convex LREE and fractionated MREE–HREE (Fig. 10a). In a multi-element diagram normalized to primitive mantle values (McDonough & Sun, 1995), most Grt websterites exhibit depletion in the highly incompatible elements (Rb, Ba) and negative Nb, Ta and Zr anomalies, but no significant anomalies in Hf and Ti. They also display negative Pb anomalies and positive Sr anomalies, except for samples BME-1 and DR10165, which have pronounced negative Sr anomalies (Fig. 10b). Samples BME-1 and DR10165 also show more obvious negative anomalies in the HFSE and Pb (Fig. 10b). Sample BM96-3 has lower contents of highly incompatible elements than other garnet pyroxenites and also shows positive anomalies in Zr and Hf but a negative Ti anomaly.

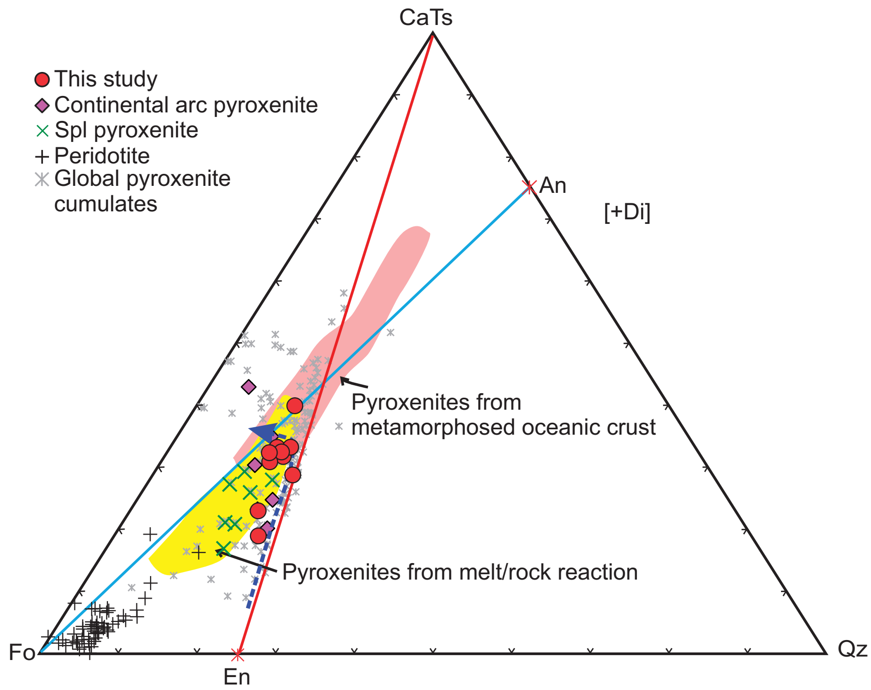

Fig. 9. Normative compositions of pyroxenite xenoliths from Lakes Bullenmerri and Gnotuk in the pseudoternary system Fo–CaTs–Qz projected from Di using the method of O'Hara (1968). Nepheline- and hypersthene-normative compositions are separated by the Di–Fo–An plane (blue line). Silica-deficient and silica-excess compositions are divided by the Di–CaTs–En plane (red line). Fo, forsterite; CaTs, Ca-Tschermak pyroxene; Qz, quartz; An, anorthite; En, enstatite; Di, diopside. The symbols and shaded fields are as in Fig. 8.

Pressure (*P*) and temperature (*T*) estimates

Thermodynamic modeling of Cpx–Opx–Grt equilibration in the system CaO–MgO–Al_2O_3–SiO_2 (CMAS; Gasparik, 2014) indicates that the primary clinopyroxene crystallized at 1420–1450°C and 2·3–2·4 GPa (Table 1), with the exception of sample BM99-5 (1460°C and 3·0 GPa; Lu *et al.*, 2017). These values are slightly higher than those estimated in volcanic systems using clinopyroxene-based thermobarometers (1324–1377°C and 2·0–2·3 GPa, respectively; Putirka, 2008). These conditions lie between the average mantle adiabats of $T_p = 1350$–1430°C (Wang *et al.*, 2002; Putirka *et al.*, 2007; Green, 2015), but lower than that expected from mantle plumes ($T_p > 1550$°C; Herzberg *et al.*, 2007; Putirka *et al.*, 2007). *P*–*T* values derived using the Fe–Mg exchange thermometer of Ellis & Green (1979) and Al-in-Opx barometer of Wood (1974) with the recrystallized mineral assemblages suggest that these xenoliths all finally equilibrated at ~970–1100°C and 1·2–1·8 GPa, thus falling along the xenolith-defined SEA geotherm (e.g. O'Reilly & Griffin, 1985). Thermobarometric estimates for the equilibration of REE between Grt and Cpx (Sun & Liang, 2015) appear to preserve evidence of an intermediate stage (1030°C and ~2·1 GPa). These *P*–*T* results indicate a decompressional cooling path (Lu *et al.*, 2017) before entrainment in the host basalt.

Sr-, Nd-, and Hf-isotope geochemistry

The Rb–Sr, Sm–Nd and Lu–Hf isotopic compositions of pyroxenitic garnet, clinopyroxene and amphibole from five samples are given in Table 6. Bulk-rock Sr-, Nd-, and Hf- isotope compositions were also reconstructed using counted mineral modes, high-precision *in situ* elemental analyses (LA-ICP-MS) and solution isotopic data on separated minerals (Table 6). Cpx has $^{87}Sr/^{86}Sr$ ranging between 0·70412 and 0·70657 (Table 6); one amphibole gives $^{87}Sr/^{86}Sr = 0·70550$, similar to that of the coexisting Cpx (~0·70566). The reconstructed bulk Sr-isotope compositions are controlled by Cpx, because of the negligible Sr contents in Grt.

The high Sr and low Rb contents of the Cpx and Amp made them suitable for *in situ* analysis. Within each sample, *in situ* Sr-isotope compositions of the megacrystalline Cpx and the discrete Cpx are indistinguishable within analytical uncertainty (Supplementary Data Electronic Appendix 2); the average data are listed in Table 7. They give $^{87}Sr/^{86}Sr$ of 0·7044–0·7084, mainly concentrated at <0·706. Compared with Cpx, amphiboles show slightly higher $^{87}Rb/^{86}Sr$ (0·008–0·083) but similar $^{87}Sr/^{86}Sr$ (0·7044–0·7087); $^{87}Sr/^{86}Sr$ values are well correlated with those of Cpx, suggesting that the Cpx and Amp have achieved near-isotopic equilibrium in terms of Sr. A marked Sr-isotope discrepancy is found in sample DR9748, where the discrete grains of Cpx have higher $^{87}Sr/^{86}Sr$ than the Cpx close to the Amp vein (0·7074 vs 0·7059). This discrepancy is also observed between the discrete Amp (~0·7069) and the vein Amp (~0·7056), which is slightly lower but less variable than those of Cpx.

Overall, Sr-isotope compositions obtained by solution are less variable than those from *in situ* analysis,

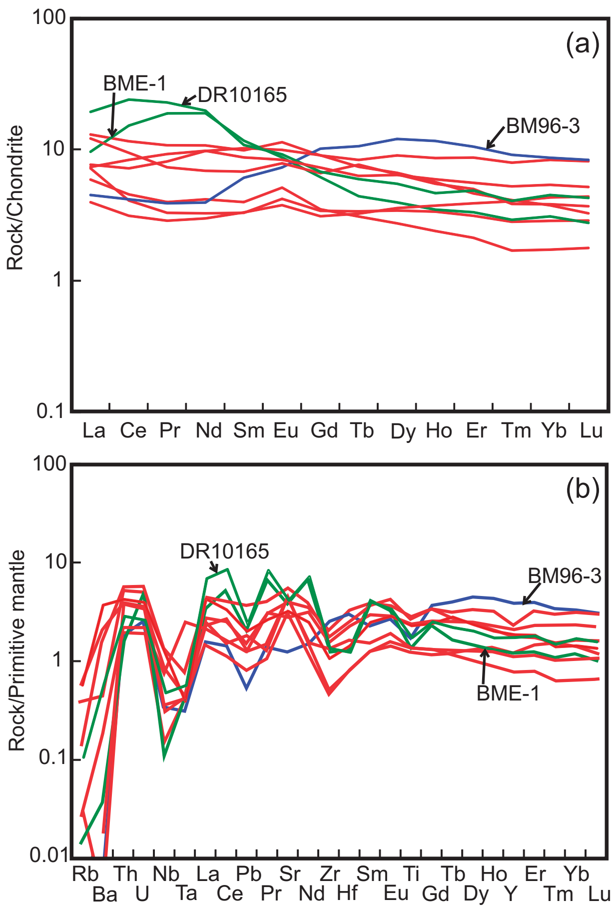

Fig. 10. Chondrite-normalized REE (a) and primitive mantle normalized trace element (b) patterns of garnet websterite xenoliths from Lakes Bullenmerri and Gnotuk. Chondrite and primitive mantle values from McDonough & Sun (1995).

but they are still positively correlated with each other, probably owing to the averaging effect of digesting whole grains.

Grt has higher $^{147}Sm/^{144}Nd$ (0·586–0·928) and more radiogenic $^{143}Nd/^{144}Nd$ (0·51255–0·51288) than the corresponding Cpx (0·117–0·210 and 0·51244–0·51274, respectively). The reconstructed bulk Nd-isotope compositions range from 0·51244 to 0·51275, essentially identical to that of Cpx; this reflects the much higher Nd and Sm contents in Cpx, and its greater modal abundance compared with Grt.

Grt has high $^{176}Lu/^{177}Hf$ (0·580–0·857) and radiogenic $^{176}Hf/^{177}Hf$ (0·28395–0·28495), whereas Cpx has a restricted $^{176}Hf/^{177}Hf$ range of 0·28281–0·28329. The reconstructed bulk Hf-isotope compositions are intermediate between those of Grt (essentially controlling Lu) and Cpx (controlling Hf), but they still show an obvious positive correlation with Cpx compositions. Internal Sm–Nd isochrons of three Grt–Cpx pairs give similar ages with a mean of ~40 Ma (Fig. 11a); Lu–Hf isochrons yield older ages ranging from 70 to 220 Ma (Fig. 11b).

Hf depleted-mantle model ages (T_{DM}) of the reconstructed bulk-rocks and clinopyroxenes are 538–682 Ma, with the exception of the reconstructed bulk-rock of BM99-4 and the clinopyroxene in BM99-2, which give negative (future) values (Table 6). Garnets have a wide range of T_{DM} from 46 to 248 Ma. In contrast, most bulk-rocks and clinopyroxenes (this study; Griffin *et al.*, 1988) give old depleted-mantle Nd model ages (mostly >1000 Ma; Table 6). However, second-stage depleted-mantle model ages (T_{DM2}), calculated assuming the absence of radiogenic ingrowth before 40 Ma (i.e. $^{147}Sm/^{144}Nd = 0$), yield an average of 388 ± 107 Ma (Table 6), comparable with the T_{DM2} ages calculated from the compositions of metasomatic peridotite xenoliths in this area (Griffin *et al.*, 1988; Stolz & Davies, 1988).

On a plot of age-corrected $^{87}Sr/^{86}Sr$ vs ε_{Nd} at $t = 40$ Ma, most samples are restricted to the field of ocean island basalt (OIB) (Fig. 12a), contrasting with the limited range of compositions reported for the host basalts ($^{87}Sr/^{86}Sr = 0·70395–0·70409$; $\varepsilon_{Nd} = +3·7$; Griffin *et al.*, 1988; Stolz & Davies, 1988). However, they overlap with the host peridotites and Cenozoic basalts, which mostly cluster within the MORB–OIB mantle array ($^{87}Sr/^{86}Sr < 0·7060$; $\varepsilon_{Nd(t)} > -4·6$), although a few peridotites metasomatized by pyroxenitic melts have $^{87}Sr/^{86}Sr$ up to 0·7105 and low $\varepsilon_{Nd(t)}$ (–7·7; Stolz & Davies, 1988). On a plot of age-corrected ε_{Nd} vs ε_{Hf} at $t = 40$ Ma, pyroxenitic clinopyroxenes are mostly scattered around the OIB field along the mantle array (Fig. 12b). Sample BM99-4 is within the OIB field, whereas sample BME-1 tends to have more radiogenic Hf at a given $\varepsilon_{Nd(t)}$. In addition, clinopyroxenes in samples BM99-2 and BM99-5 have similar Nd-isotope compositions but very different $\varepsilon_{Hf(t)}$ (+18·6 and +1·69, respectively).

DISCUSSION

Origin of the garnet pyroxenites

Pyroxenites are an important component in the Earth's upper mantle, yet their genesis remains controversial. Proposed pyroxenite-forming processes are variable, as pyroxenites are petrologically, chemically and isotopically diverse (e.g. Downes, 2007). We here discuss the origin of the garnet websterites as potentially representing (1) solidified melts, (2) melt–rock reaction products, (3) metamorphosed oceanic crust, or (4) cumulates. If the pyroxenites represent solidified melts, their bulk-rock compositions would be similar to those of primary melts generated by partial melting of peridotite, or more evolved melts. However, their MgO contents are too high to represent typical basaltic melts at mantle depths. In particular, melts with such high MgO contents (i.e. 17·0–20·8 wt %) have been produced experimentally by melting dry fertile mantle compositions at ~1500–1600°C (e.g. Falloon *et al.*, 1988; Hirose & Kushiro, 1993; Walter, 1998); such melts are associated with Phanerozoic mantle plumes (i.e. Hawaii picrites) or the Archean subcontinental mantle (e.g. komatiites). Picrites or komatiites have either accumulated olivine or are olivine-saturated, even at high pressure, so that

Table 6: Sr–Nd–Hf isotopes of clinopyroxene, amphibole, garnet and whole-rock of garnet pyroxenites from Lakes Bullenmerri and Gnotuk

Sample		Rb (ppm)	Sr (ppm)	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$ (±2SE)	Sm (ppm)	Nd (ppm)	$^{147}Sm/^{144}Nd$	$^{143}Nd/^{144}Nd$ (±2SE)	ε_{Nd} (t = 40 Ma)	T_{DM} (Ma)	T_{DM2} (Ma)	Lu (ppm)	Hf (ppm)	$^{176}Lu/^{177}Hf$	$^{176}Hf/^{177}Hf$ (±2SE)	ε_{Hf} (t = 40 Ma)	T_{DM} (Ma)
BHVO-2					0·703468±8				0·512985±3							0·283107±10		
BM99-4	Cpx	0·04	149	0·0008	0·705659±3	2·25	7·93	0·179	0·512607±3	–0·32	2361	420	0·03	1·14	0·004	0·282901±3	4·87	538
	Amp	1·57	493	0·0090	0·705500±5	2·78	9·97	0·176	0·512699±3	1·48	1816	354	0·04	0·72	0·008	0·282959±7	6·85	504
	Grt					0·56	0·38	0·928	0·512846±5	0·47		391	1·13	0·18	0·857	0·283947±6	19·3	46
	WR*					1·48	4·88	0·191	0·512611±3	–0·34	3528	421	0·20	0·71	0·040	0·282945±3	5·49	—
BME-1	Cpx		127		0·705815±14	2·82	15·22	0·117	0·512436±4	–3·38	1123	532		0·51	0·002	0·283051±5	10·2	
	Cpx-R		127		0·705848±3				0·512442±3							0·283072±26	11·0	
	Grt					0·61	0·66	0·586	0·512545±12	–3·64		542	0·35	0·09	0·580	0·284281±8	38·5	102
	WR*					1·77	8·71	0·128	0·512444±2	–3·28	1252	528	0·07	0·37	0·026	0·283107±5	11·6	629
BM99-2	Cpx		114		0·704123±4	0·72	2·42	0·188	0·512833±8	4·00	1894	262	0·01	0·30	0·006	0·283290±11	18·6	—
	Grt												0·27	0·09	0·404	0·284950±21	66·8	248
	WR*												0·11	0·23	0·067	0·283507±11	26·0	555
BM99-5	Cpx		109		0·705659±3	2·23	6·69	0·210	0·512862±6	4·46	11683	245	0·04	1·38	0·004	0·282811±8	1·69	682
DR9748†	Cpx	0·013	116	0·0003	0·70657±4	2·50	9·00	0·169	0·512738±5	2·25	1391	326						
	Opx					0·29	0·94	0·185	0·512748±6	2·36	2159	322						
	Grt					0·79	0·57	0·833	0·512876±50	1·55		352						
	WR					1·38	4·89	0·171	0·512745±16	2·37	1451	322						
GN35†	WR	0·29	57·6	0·0146	0·704935±16	0·92	2·76	0·202	0·512838±6	3·87	4069	261						
DR10165†	WR	0·36	85	0·0123	0·708579±10	1·62	8·91	0·110	0·512413±20	–3·95	1088	547						

WR, whole-rock; Cpx-R, replication clinopyroxene; T_{DM} and T_{DM2}, single- and second-stage depleted mantle model ages, respectively. Accepted decay constants ($\lambda_{Sm} = 6{\cdot}54 \times 10^{-12}\,a^{-1}$ and $\lambda_{Lu} = 1{\cdot}865 \times 10^{-11}\,a^{-1}$) were used. The present-day chondritic uniform reservoir (CHUR) values of $^{147}Sm/^{144}Nd = 0{\cdot}1960$, $^{143}Nd/^{144}Nd = 0{\cdot}512630$ and $^{176}Lu/^{177}Hf = 0{\cdot}0336$, $^{176}Hf/^{177}Hf = 0{\cdot}282785$ are taken from Bouvier *et al.* (2008); the present-day depleted mantle values of $^{147}Sm/^{144}Nd = 0{\cdot}21357$, $^{143}Nd/^{144}Nd = 0{\cdot}51315$ and $^{176}Lu/^{177}Hf = 0{\cdot}0384$, $^{176}Hf/^{177}Hf = 0{\cdot}283251$ are from DePaolo (1981) and Griffin *et al.* (2000), respectively.

*Whole-rock Sm–Nd and Lu–Hf isotopic compositions were reconstructed from Cpx and Grt, which dominate the Sm, Nd, Lu and Hf budget.

†Data from Griffin *et al.* (1988).

they can reach such high MgO and Mg# (Lee *et al.*, 2006). In the present case, the low Ni contents (<500 ppm) and high CaO (Supplementary Data Electronic Appendix 1) contents of the garnet pyroxenites preclude significant olivine accumulation. Experimental studies suggest that the addition of H_2O reduces the solidus temperature by 100–150°C compared with anhydrous melting of a lherzolite source, which means that the pyroxenites could alternatively come from solidified magmas derived from the flux melting of a fertile mantle source at ~1380–1480°C beneath volcanic arcs (Ulmer, 2001). However, the pyroxenites have Mg# higher than any type of primitive arc magmas (Kelemen *et al.*, 2003*a*). The garnet pyroxenites are therefore unlikely to represent solidified basaltic melts.

Table 7: Mean Sr isotopic compositions of clinopyroxene and amphibole in the garnet pyroxenite from Lakes Bullenmerri and Gnotuk by LA-MC-ICP-MS

Sample	Mineral	Rb	Sr	$^{87}Rb/^{86}Sr$	$^{87}Sr/^{86}Sr$	1σ
BME-1	Cpx (5)		127		0·70610	0·00014
	Amp (4)	1·01	371	0·008	0·70605	0·00007
BM99-4	Cpx (7)	0·04	149	0·001	0·70627	0·00008
	Amp (4)	1·57	493	0·009	0·70604	0·00004
BM99-2	Cpx (4)		114		0·70447	0·00008
BM99-5	Cpx (4)		109		0·70685	0·00016
	Amp (4)	8·88	384	0·068	0·70612	0·00007
BM96-3	Cpx (11)		104		0·70621	0·00009
	Amp (3)	3·17	359	0·026	0·70599	0·00003
BM96-2	Cpx (5)		146		0·70444	0·00007
	Amp (5)	3·17	359	0·067	0·70443	0·00004
BM99-6	Cpx (11)		137		0·70449	0·00006
	Amp (4)	6·25	386	0·048	0·70461	0·00006
DR10165	Cpx (6)		136		0·70844	0·00081
	Amp (6)	2·31	540	0·012	0·70872	0·00001
GN35	Cpx (5)		96		0·70563	0·00045
	Amp (4)	8·77	311	0·083	0·70549	0·00009
DR9748	Cpx (4)		135		0·70749	0·00022
	Cpx-V (4)				0·70598	0·00026
	Amp (4)	3·88	525	0·022	0·70698	0·00004
	Amp-V (4)	13·06	577	0·066	0·70569	0·00001

Cpx-V, Amp-V, clinopyroxene and amphibole in the vein, respectively. Numbers in parentheses indicate the number of analyzed grains.

The second hypothesis is that the garnet pyroxenites are the products of melt–rock reaction. It has been shown on theoretical, observational and experimental grounds that pyroxene-rich mantle assemblages can be generated by reaction of a silicic melt with peridotite (Garrido & Bodinier, 1999; Rapp *et al.*, 1999; Liu *et al.*, 2005; Bodinier *et al.*, 2008; Lambart *et al.*, 2012). Such 'secondary' pyroxenites have compositions transitional between peridotites and silicic mantle melts, and thus plot between the En–CaTs and Fo–An joins in the CaTs–Fo–Qz diagram projected from Di (Lambart *et al.*, 2012; Mallik & Dasgupta, 2012; Tilhac *et al.*, 2016). They also commonly have high Mg#, and high concentrations of Ni and other compatible elements released by olivine when reaction with a silicic melt converts olivine to pyroxene to generate pyroxenites or pyroxene-rich peridotites (Garrido & Bodinier, 1999; Liu *et al.*, 2005; Tilhac *et al.*, 2016). These features are shown by the spinel pyroxenites from Lakes Bullenmerri and Gnotuk, which have high Mg# (up to 90; Fig. 8) and Ni contents (760–924 ppm; Griffin *et al.*, 1984). In addition, composite xenoliths of spinel pyroxenites and their host lherzolites accordingly show variable Cr_2O_3 in spinel across the contact zone (Fig. 5f; Griffin *et al.*, 1984). However, the garnet pyroxenites studied here lie mostly along the En–CaTs join (Fig. 9) and show a limited range of low Ni contents. As noted above, the garnet pyroxenites provide little or no compositional evidence for the involvement of olivine in their formation, indicating limited, if any, contribution of melt–peridotite reactions.

Recycling of crustal material as a mechanism for the formation of garnet pyroxenites has attracted much attention [see reviews by Spray (1989) and Downes (2007)]. This type of pyroxenite is expected to inherit its chemical and isotopic compositions from low-*P* crustal cumulates (e.g. gabbro and MORB). These pyroxenites would have lower Mg# and MgO but higher Al_2O_3, CaO and Na_2O contents than most mantle pyroxenites

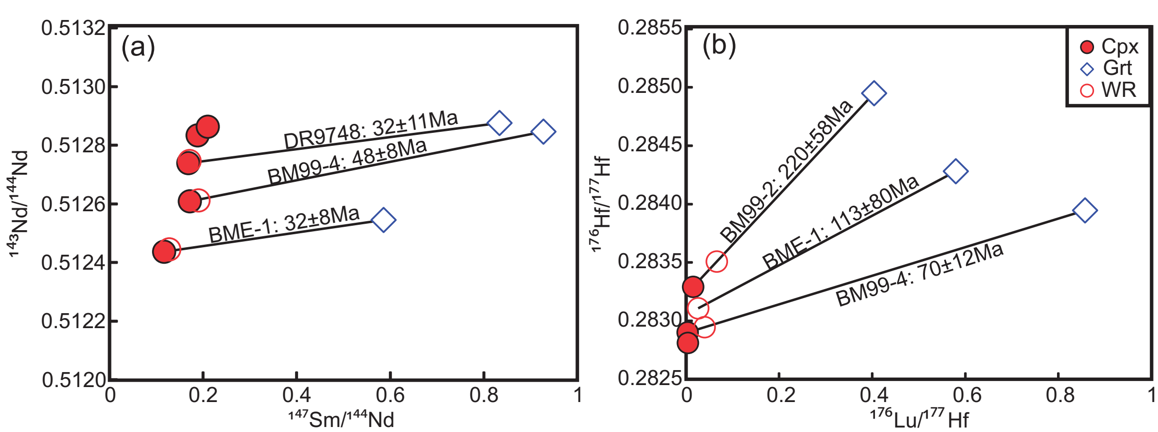

Fig. 11. Sm–Nd (a) and Lu–Hf (b) internal isochrons defined by Grt and Cpx pairs in the garnet pyroxenites from Lakes Bullenmerri and Gnotuk. Ages are reported at 2σ level.

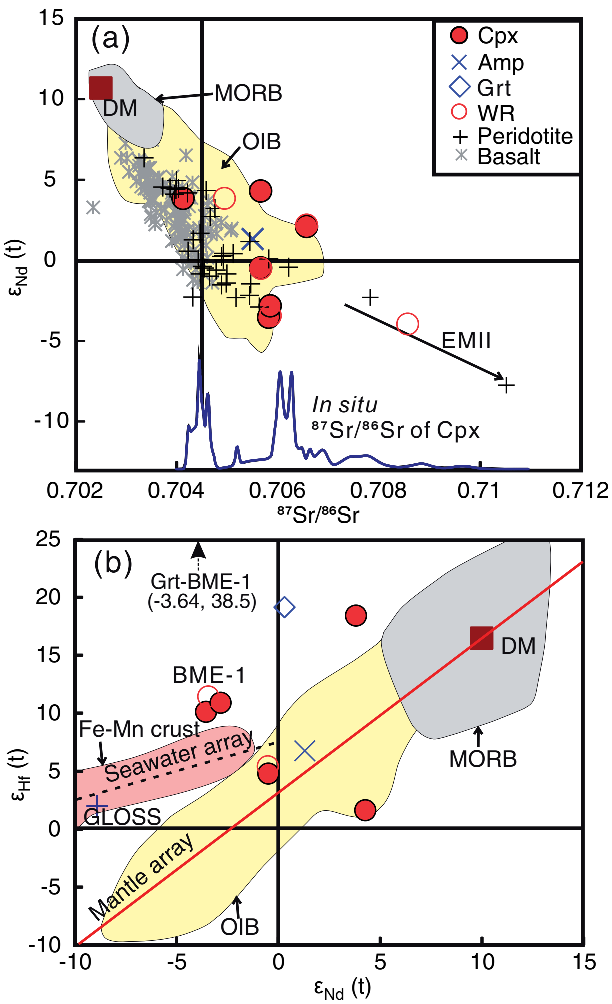

Fig. 12. Variation of $^{87}Sr/^{86}Sr$ vs ε_{Nd} (a) and ε_{Nd} vs ε_{Hf} (b) at $t = 40$ Ma for Cpx, Amp and whole-rocks (WR) of garnet pyroxenites from Lakes Bullenmerri and Gnotuk. In (a), fields for MORB and OIB are from Zindler & Hart (1986); data for peridotites from this area are from Griffin *et al.* (1988) and Stolz & Davies (1988); data for Cenozoic basalts in easternAustralia are from O'Reilly & Zhang (1995), Vogel & Keays (1997), Zhang & O'Reilly (1997), McBride *et al.* (2001), Demidjuk *et al.* (2007), Price *et al.* (2014) and Oostingh *et al.* (2016). In (b), fields for MORB, OIB and the mantle array are from Vervoort *et al.* (1999); field for Fe–Mn crust and seawater array are from Albarède *et al.* (1998).

(Fig. 8), and would lie along the Fo–An join in the Fo–CaTs–Qz (+Di) system (Fig. 9). Their whole-rock trace element patterns also may show positive Eu and Sr anomalies, relatively low HREE and MREE, and high LREE/HREE owing to the accumulation of primary plagioclase. In addition, garnet and clinopyroxene (the two dominant phases) show 'ghost' signatures of plagioclase (positive Eu and Sr anomalies), and garnet has depleted HREE patterns, suggesting a metamorphic origin (Morishita *et al.*, 2009; Yu *et al.*, 2010; Svojtka *et al.*, 2016). Despite their positive Eu anomalies, the garnet pyroxenites studied here are clearly distinct from recycled oceanic gabbros and primary MORB (Figs 8 and 9), precluding their origin as recycled mafic crust in the convective mantle.

Their reconstructed bulk-rock compositions are indeed similar to those of other pyroxenites recognized as cumulates worldwide (Figs 8 and 9), and they plot among the pyroxenitic cumulates produced experimentally from high-*P* hydrous basalts (Müntener *et al.*, 2001). Therefore, the most probable hypothesis for the origin of the garnet pyroxenites is that they are Cpx-dominated cumulates (*sensu lato*) from relatively primitive mantle melts (Bultitude & Green, 1971).

Nature of the parental melt(s) and source region

To clarify the geochemical affinity of the parental melt(s) of the garnet pyroxenites, we have calculated the melt compositions in equilibrium with the reconstructed primary clinopyroxenes (Supplementary Data Electronic Appendix 3). We used partition coefficients reported by Bodinier *et al.* (1987) for Cpx at 1400°C and 1·5–2·5 GPa, comparable with our thermobarometric estimates for the initial crystallization conditions, and the procedure of Hanson & Langmuir (1978). Calculated melts have compositions typical of tholeiitic basalts (Supplementary Data Electronic Appendix 3), as indicated by their SiO_2 contents (~50 wt %) and the variations in Mg# (58–64). The melt in equilibrium with the reconstructed Cpx for sample BM99-5 has high MgO (~10·9%) but lower Mg# (~48·1), probably indicating Fe enrichment in the source, which is reflected in the exsolution of ilmenite from clinopyroxene megacrysts during cooling. Sample BM99-4 contains primary orthopyroxene, and thus probably represents the most primitive pyroxenite in this study, as Opx is expected to appear early on the liquid line of descent of tholeiites (Demarchi *et al.*, 2001).

We also have estimated the trace element compositions of melts in equilibrium with the reconstructed primary Cpx (Fig. 13; Supplementary Data Electronic Appendix 3). The partition coefficients used in the calculations are taken from Adam & Green (2006) except for Ba, Sr, Eu, Dy and Er, which are from Hart & Dunn (1993). Calculated melts for samples BM99-4 and BM99-5 show tholeiite-like or OIB-like trace element characteristics, with moderate enrichment of LREE and the absence of HFSE anomalies (Fig. 13). These garnet websterites also show OIB-like Sr-, Nd-, and Hf-isotope compositions (Fig. 12). Their melts may represent the primary compositions of magmas parental to garnet pyroxenites. For sample BM99-2, calculated melts have REE patterns similar to those of sample BM99-4 but lower REE contents, probably reflecting a differentiation process. Therefore, it seems that the garnet pyroxenites studied here crystallized from primitive to moderately differentiated tholeiitic basalts.

However, experimental studies performed at 2 and 3 GPa on a typical anhydrous tholeiitic basalt (MORB-like eclogite G2; Pertermann & Hirschmann, 2003*a*, 2003*b*), suggest that only extremely high degrees of melting (>90%, or, equivalently, extremely low degrees of crystallization from a magma of this composition)

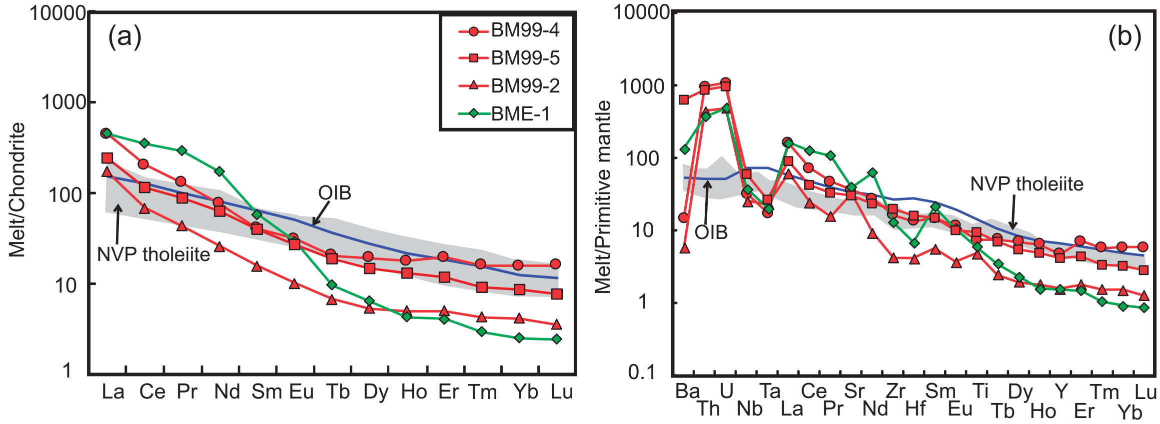

Fig. 13. REE patterns normalized to chondrites (a) and trace element diagrams normalized to primitive mantle (b) for melts in equilibrium with garnet pyroxenites from Lakes Bullenmerri and Gnotuk. Patterns are calculated using the reconstructed Cpx and mineral–melt partition coefficients listed by Hart & Dunn (1993) and Adam *et al.* (2006). The data for Newer Volcanic Province (NVP) tholeiitic basalts are from Price *et al.* (1997), Vogel & Keays (1997) and McBride *et al.* (2001). The data for OIB are from McDonough & Sun (1995).

could produce the high MgO values characteristic of the garnet pyroxenites studied here (Fig. 14a). Moreover, the garnet pyroxenites tend to have lower CaO and Al_2O_3 than the melting residues (Fig. 14b–d). It thus seems unlikely that the garnet pyroxenites were derived from anhydrous tholeiitic basalts.

The garnet pyroxenites have compositions similar to those of the high-MgO Sierra Nevada arc pyroxenites, which are interpreted as cumulates from mantle-wedge-derived hydrous basalt or basaltic andesite (Müntener *et al.*, 2001; Lee *et al.*, 2006). Müntener *et al.* (2001) showed that a hydrous high-MgO basaltic andesite (51·7 wt % SiO_2, 16·4 wt % Al_2O_3, 10·8 wt % MgO, 9·67 wt % CaO and Mg# = 71; Baker *et al.*, 1994) could crystallize clinopyroxene–orthopyroxene (± garnet) assemblages at 1·2 GPa. In terms of major elements (Fig. 14), the garnet pyroxenites studied here are closely similar to experimental cumulates from a hydrous basaltic andesite (Fig. 14). Furthermore, discrete amphibole grains with minor Cl contents (Table 6) are abundant in the garnet pyroxenites, and they are in equilibrium with the recrystallized Cpx in the garnet pyroxenites (Fig. 3e), consistent with crystallization of amphibole from the migrating or trapped residual melts (Harte *et al.*, 1993). Minor 'exsolution-like' amphibole lamellae and pods (Fig. 3d) and F–Cl-bearing apatite inclusions were also found in the clinopyroxene megacrysts. These results indicate a high volatile budget in the bulk system. In addition, these xenoliths also contain abundant fluid inclusions and cavities (>3 vol. %) trapped at high pressure (Andersen *et al.*, 1984; Griffin *et al.*, 1984; O'Reilly *et al.*, 1990). CO_2 is the dominant phase now present, but it is interpreted as the residual fluid left after a post-entrapment reaction from an initial CO_2–H_2O mixture, which contained small amounts of halogens (F, Cl) and sulfur species (Andersen *et al.*, 1984). The occurrence of halogens may suggest an oxidizing environment. We thus suggest that upon percolation, these volatiles were released into the surrounding wall-rock and most of the H_2O reacted to produce amphibole, leaving a CO_2-rich fluid. Such a compositional shift from H_2O-rich fluids at depth to CO_2-rich residual liquids at shallower levels, and associated amphibolization, have also been described in experimental results (Schneider & Eggler, 1986) and in other arc-related pyroxenites (Burg *et al.*, 1998; Tilhac *et al.*, 2016).

A mantle-wedge environment is specifically suggested for the genesis of the parental melts. Samples BME-1 and DR10165 show strong enrichment of large ion lithophile elements (LILE) and LREE and marked negative anomalies of HFSE, a typical arc signature (Fig. 10). Furthermore, clinopyroxene and the bulk-rocks in BME-1 and DR10165 have radiogenic Sr-isotope compositions ($^{87}Sr/^{86}Sr$ up to 0·708) and low $\varepsilon_{Nd(t)}$, but relatively high $\varepsilon_{Hf(t)}$, which plot between a depleted-mantle source and a subducted oceanic sediment (Fig. 12). Trace element compositions and decoupled Nd–Hf isotopes probably suggest that these garnet pyroxenites were derived from a metasomatic mantle source contaminated by subduction-related sediments (e.g. Griffin *et al.*, 1988). The melt in equilibrium with the reconstructed Cpx of BME-1 also has highly fractionated LREE/HREE and depleted HFSE (Fig. 13), and a high Sr/Y ratio, similar to the melt produced by metasomatized peridotite (e.g. Rapp *et al.*, 1999). In addition, the estimated crystallization temperature of the primary Cpx (1420–1450°C and 2·3–2·3 GPa) is close to the mantle potential temperature (Table 1), indicating that the melts parental to the garnet pyroxenites studied here did not form in the lithospheric mantle but in the hottest part of the mantle wedge (1350–1450°C, 2·0–2·5 GPa; Ulmer, 2001; Kelemen *et al.*, 2003*b*), where tholeiitic or high-Al basaltic melts were produced during the initial stage by high degrees of partial melting (10–15%) of a mantle source with a low water content (<0·0 wt %; Baker *et al.*, 1994; Ulmer, 2001; Grove *et al.*, 2006; Kushiro, 2007). Such low H_2O contents in the initial lavas are found in the Cascades,

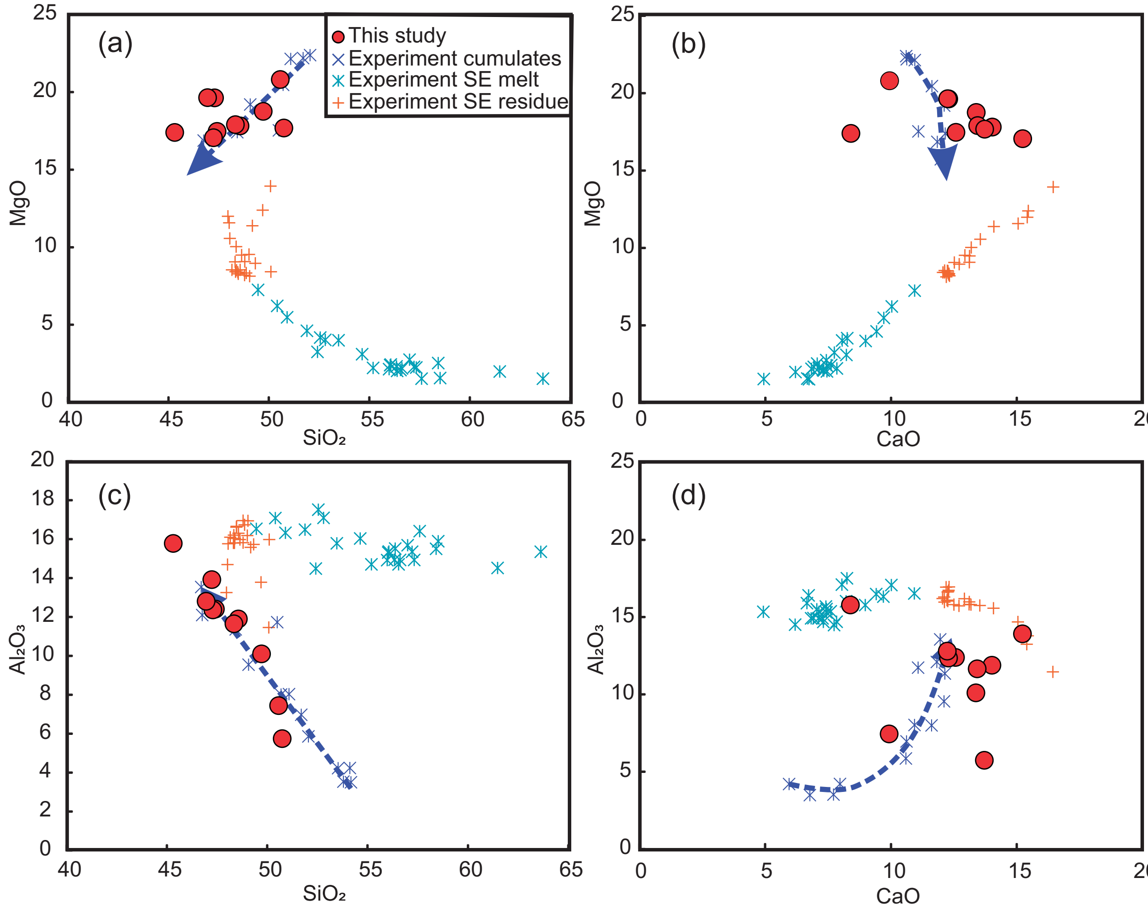

Fig. 14. Covariation diagrams modelling petrogenetic relationships between garnet pyroxenites (this study), experimental cumulates from hydrous basaltic andesite at 1·2 Ga (Müntener *et al.*, 2001), melt compositions and solid residues from partial melting of anhydrous MORB-like eclogite (similar to a typical anhydrous tholeiitic basalt; Pertermann & Hirschmann, 2003*b*).

western USA and Galunggung, Indonesia (Sisson & Layne, 1993; Sisson & Bronto, 1998). We thus propose that the garnet pyroxenites may best be interpreted as cumulates from tholeiitic magmas produced by hydrous melting of metasomatized fertile peridotite in a convective mantle wedge.

Constraints on the age of the garnet pyroxenites

No zircons or other phases directly datable by the U–Pb system were found in these pyroxenite xenoliths and thus ages could only be estimated from isochron and model ages calculated from our radiogenic isotopic data. Generally, the Lu–Hf system has a higher closure temperature for Cpx–Grt pairs (Blichert-Toft, 1999; Cheng *et al.*, 2008; Gonzaga *et al.*, 2010; Shu *et al.*, 2014); it thus resists disturbance during thermo-tectonic processes, and may preserve primary age information (Jacob *et al.*, 2004; Gonzaga, 2007; Xiong *et al.*, 2015). The Rb–Sr and Sm–Nd systems both have lower closure temperatures and are more easily disturbed by later events, such as the interaction with fluids (e.g. Gysi *et al.*, 2011; Xiong *et al.*, 2014; Tilhac *et al.*, 2017).

Model ages theoretically correspond to the timing of separation (e.g. change of parent–daughter ratios following melt extraction) of a given sample (or domain) from a modelled reservoir, for instance, a convecting upper mantle (depleted mantle source). Depleted-mantle Nd model ages (T_{DM}) of the garnet pyroxenites are over 1000 Ma (Table 6; Griffin *et al.*, 1988), and older than those of peridotite xenoliths from these localities that were metasomatized by pyroxenitic melts (concentrated between 600 and 1000 Ma; Griffin *et al.*, 1988; Stolz & Davies, 1988). This may suggest that the Sm–Nd system in the garnet pyroxenites has been disturbed below its blocking temperature by later events, possibly the infiltration of volatile-rich fluids (leading to the formation of vein amphibole and phlogopite; Figs 3f and 4e, f) and thus Nd model ages could be meaningless. However, second-stage Nd model ages (T_{DM2}) calculated from these garnet websterites are comparable with the estimates (300–500 Ma; Griffin *et al.*, 1988) derived from a mixing model based on the Sr and Nd isotopic compositions of metasomatic peridotites and melt-related garnet pyroxenites, and may represent the intrusive ages of the garnet pyroxenites (Griffin *et al.*, 1988). One garnet pyroxenite from the studied area has given a whole-rock Re–Os model age (T_{MA}) of 560 Ma (McBride *et al.*, 1996) and clinopyroxenes and reconstructed bulk-rocks in this study give Hf model ages of 538–682 Ma (Table 6). Considering the geological history of the area, it is reasonable to conclude that the cumulate precursors of the garnet pyroxenites could have crystallized at ~600–300 Ma during the extended Paleozoic orogenic history of the area, although we emphasize that these model ages have large uncertainties.

Garnet–clinopyroxene pairs give similar Sm–Nd isochron ages with a mean of 40 Ma (Fig. 11a). If this age is meaningful, it would represent a 'cooling age' below the blocking temperature of the system rather than the actual timing of pyroxenite formation (Blichert-Toft, 1999; Montanini *et al.*, 2006, 2012; Ishikawa *et al.*, 2007; Gysi *et al.*, 2011; Shu *et al.*, 2014; Xiong *et al.*, 2014; Borghini *et al.*, 2016). Petrographic observations suggest that most garnets have formed by subsolidus exsolution from clinopyroxene megacrysts upon cooling and reconstructed bulk-rock trace element compositions typically show flattened patterns in the MREE–HREE range, indicating little or no contribution of primary garnet to their whole-rock compositions. However, internal Lu–Hf mineral isochrons yield a range of ages older than the Sm–Nd isochron ages (Fig. 11b), suggesting the incomplete resetting of Lu–Hf mineral isochrons during cooling. Le Roux *et al.* (2016) suggested that Lu/Hf ratios are more uncertain as they were probably affected by subsolidus re-equilibration even in clinopyroxene, as opposed to Nd and Sm, for which subsolidus re-equilibration is barely detectable in clinopyroxene. This may reflect diffusion-related effects during cooling, as radiogenic Hf diffuses out of garnet more slowly than does radiogenic Nd (Duchene *et al.*, 1997). Nonetheless, the Sm–Nd system is more easily disturbed by interaction with fluids; percolation of fluid or melt drives the system closer to final equilibrium and increases the length and time scales for diffusive re-equilibration in the system (Liang, 2014). Three Grt websterites contain abundant Amp, and it can therefore be surmised that the recrystallized clinopyroxene and exsolved garnet define an Eocene cooling age of ~40 Ma (Cpx–Grt Sm–Nd isochrons), clearly predating the xenolith entrapment by Cenozoic basalts.

Tectonic implications

The regional geological context and geochronological studies of SE Australia reveal that a series of convergent-margin orogens developed along the southern or eastern margins of the Gondwanan supercontinent from the late Neoproterozoic (*c.* 700 Ma) to early Mesozoic (Coney *et al.*, 1990; Cawood, 2005). They record a cycle of continental rifting and ocean opening, subduction (starting *c.* 514 Ma), arc–continent collision, and important post-collisional extension and magmatism with exhumation of the underthrust passive margin in the Delamerian fold belt (Fig. 1a; see review by Crawford *et al.*, 2003) owing to rollback of the proto-Pacific slab. Then the subduction zone jumped outboard and formed new subduction systems during the early Ordovician to late Devonian in the Lachlan fold belt and from Devonian to Mesozoic time in the New England fold belt (Fig. 1a; VandenBerg *et al.*, 2000). Each cycle, encompassing sedimentation, volcanism and granite intrusion, ended with an orogeny (Glen, 2013). This episodic convergent tectonics suggests that present-day SE Australia was located within the continental-margin arcs or back-arc regions where the mantle wedge could be metasomatized by slab-derived fluids and/or melts, and could produce melts parental to the garnet pyroxenites. The temperature–pressure estimates for the primary clinopyroxene crystallization are 1420–1450°C at 2·3–2·4 GPa (Stage A in Fig. 15), corresponding to the melting conditions in the hottest part of a mantle wedge according to multiple saturation experiments (1350–1450°C, 2·0–2·5 GPa; Ulmer, 2001) and numerical models (Kelemen *et al.*, 2003*b*; England & Katz, 2010).

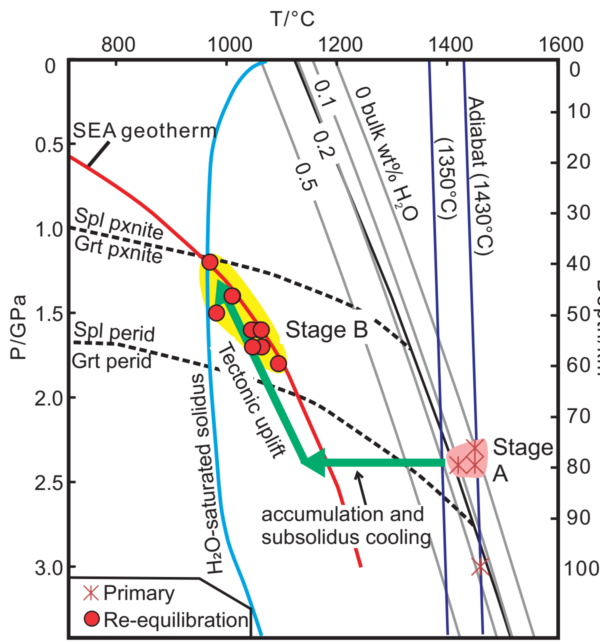

Fig. 15. *P–T–t* diagram illustrating the thermal evolution of garnet websterite from Lakes Bullenmerri and Gnotuk (modified from Lu *et al.*, 2017). Dry (thin black line) and water-saturated (thick blue line) solidi for fertile peridotites are from Hirschmann (2000) and Katz *et al.* (2003). Solidi calculated using the parameterization of Katz *et al.* (2003) after 10% partial melting of dry peridotite, with subsequent addition of variable amounts of water (up to 0·5 bulk wt %; grey lines). Average current mantle adiabat is shown for potential temperature of 1350°C and 1430°C (Putirka, *et al.*, 2007; Green, 2015). The spinel pyroxenite (Spl pxnite)–garnet pyroxenite (Grt pxnite) boundary is from Herzberg (1978); the spinel peridotite (Spl perid)–garnet peridotite (Grt perid) boundary is from Gasparik (2014). Southeastern Australian (SEA) xenolith-based geotherm is from O'Reilly & Griffin (1985). Red shaded area (Stage A) corresponds to primary pyroxene crystallization from hydrous tholeiitic magma generated in the hottest region of the mantle wedge by high melting degrees (10–15%) at relatively low water contents (<0·2 wt % in the mantle source). Yellow shaded area (Stage B) indicates the stability field of the garnet websterites before entrainment in Cenozoic basalts. Green arrow shows a possible *P–T* path from initial crystallization (Stage A) to final re-equilibration conditions (Stage B).

Combined with the previous discussion, we therefore suggest that the parental magmas of the garnet pyroxenites formed by partial melting of a hydrous mantle wedge during Paleozoic accretion orogeny cycles.

From the late Jurassic to the early Cretaceous, Australia drifted slowly northwards from Antarctica forming a series of rift basins that accumulated thicknesses of many kilometres of non-marine fluvial sediments (Edwards *et al.*, 1996; Holdgate & Gallagher, 2003). The spreading led to the opening of the Tasman Sea (~95 Ma) and presumably to the uplift of the marginal subcontinental lithospheric mantle (Veevers, 1986). From *c.* 40 Ma, the spreading rate increased and stabilized, such that the Australian continent was completely separated from the Antarctic plate (Tikku & Cande, 1999; McDougall, 2008). This is also consistent with regional uplift and the formation of rift basins (i.e. Cull *et al.*, 1991; Edwards *et al.*, 1996; O'Reilly *et al.*, 1997; Betts *et al.*, 2002). The 40 Ma ages correspond to the internal Cpx–Grt Sm–Nd isochrons calculated for the garnet pyroxenites in this study, which thus may be related to the regional uplift and extension events. This interpretation is supported by the *P*–*T* of 1·2–1·8 GPa (~40–60 km) and 950–1100°C for the re-equilibrated garnet pyroxenites, consistent with cooling during an uplift of at least 10–20 km, long before they were entrained in the host basalts (Fig. 15).

CONCLUSIONS

Petrographic observations and geochemical data suggest that the garnet pyroxenite xenoliths in the Lake Bullenmerri and Gnotuk maars experienced multiple equilibration and metasomatic events before being entrained by the host basanite. In the first stage, clinopyroxene-dominant cumulates with minor spinel and orthopyroxene precipitated from hydrous tholeiitic magmas at about 75 km depth in the mantle. These melts were derived from partial melting of a convective mantle wedge infiltrated by fluids from a subducted slab during the Paleozoic accretionary orogenic cycles mapped in the overlying crust. The melts intruded the lithospheric mantle, acted as metasomatic agents and crystallized megacrystic clinopyroxenites, the protoliths of the garnet websterites.

Subsequently (at ~40 Ma), the clinopyroxenite cumulates re-equilibrated by extensive exsolution of garnet and orthopyroxene (± ilmenite), which then variably recrystallized during cooling to the ambient geotherm at ~40–60 km. The garnet pyroxenites reveal a decompressional cooling path; they and their host peridotites were tectonically uplifted to shallower level during Cenozoic rifting associated with crustal inflation and thinning in southeastern Australia. The xenolith suite thus remarkably records a long history of heating, magmatism, cooling and uplift in the uppermost subcontinental mantle; it represents a rare case in which mantle events recorded in xenoliths can be closely tied to vertical tectonics in the overlying crust.

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SUPPLEMENTARY DATA

Supplementary data for this paper are available at *Journal of Petrology* online.

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